

Partons, QCD, and the Inner Life of the Proton

A Comprehensive Reading Companion for Particle Physics

Prepared for Parker

Covering Griffiths *Introduction to Elementary Particles*, 1st ed., §8.3–8.6
and Halzen & Martin *Quarks and Leptons*, §10.1–10.6, 11.2

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How to read this book

This document is a self-contained reading companion written specifically for the deep-inelastic-scattering / parton model / QCD unit of your particle physics course. It is organised to parallel two canonical textbooks of the subject: David Griffiths’ *Introduction to Elementary Particles* (1st edition) for sections 8.3–8.6, and Francis Halzen and Alan Martin’s *Quarks and Leptons: An Introductory Course in Modern Particle Physics* for sections 10.1–10.6 and 11.2.

The two books tell the same physical story, but with a slightly different emphasis. Griffiths is warm, conversational, and deliberately sparing with field-theoretic technology: he wants you to see the picture, and he does not mind waving his hands a little bit along the way. Halzen and Martin (henceforth *H&M*) are a little more formal; they write down more amplitudes and do more algebra, and they are more careful about the distinction between hadron-level and parton-level kinematics. You benefit from reading both.

Organisation. The document has four parts:

Part I (Chapters 1–4) follows Griffiths §8.3–8.6. We start from the Mott cross section, climb through elastic electron–proton scattering and the Rosenbluth formula, turn to inelastic scattering and the two structure functions $W_{1,2}$, and meet Bjorken scaling and Feynman’s parton model. The part ends with an accounting of valence and sea quarks inside the proton.

Part II (Chapters 5–9) follows H&M §10.1–10.6. Here we include gluons: first as force carriers, then as partons. We rewrite the parton model in “hatted” partonic variables, compute the first QCD correction from gluon emission, derive the Altarelli–Parisi (DGLAP) splitting functions, and assemble the evolution equations that replace exact Bjorken scaling with its logarithmic cousin.

Part III (Chapter 11) follows H&M §11.2. Parton distribution functions describe what goes *into* a hadron; fragmentation functions describe what comes *out*. They obey the same DGLAP equations with the same splitting kernels, and the two halves of the parton model fit together in semi-inclusive processes.

Part IV (Chapter 12) steps back, pulls the main threads together, and gives you a road map for the HW 5 problem set.

Features. Each chapter is broken into short sections, each devoted to one physical idea. Derivations are presented in enough detail that you can verify them line by line. Sprinkled throughout are coloured boxes: *Key ideas* state the thing you should carry with you, *Margin notes* make an aside that is worth pausing on, *Worked examples* do a concrete calculation, and *Self-check* questions ask you to do one yourself. Many figures are drawn with TIKZ from scratch and should be zoomable in a modern PDF reader.

Notation and units. We use natural units $\hbar = c = 1$ throughout, except in a few places where restoring $\hbar c = 197.3 \text{ MeV} \cdot \text{fm}$ makes the physics clearer. Four-vectors are written in the “mostly-minus” metric $g^{\mu\nu} = \text{diag}(+, -, -, -)$, so that $p^2 = m^2$ for an on-shell particle and $q^2 < 0$ for a space-like virtual photon. We define $Q^2 \equiv -q^2 > 0$. Repeated Lorentz indices are summed, and Dirac matrices follow the conventions of Peskin and Schroeder (which happen to coincide with H&M’s).

A reading strategy. On a first pass, read each chapter from start to finish, but skim the longer derivations. Pay attention to the figures and the coloured boxes; these are the “spine” of each chapter. On a second pass, work the self-check questions and fill in the derivations line by line. If you are pressed for time, the absolute minimum is: Chapter 3 (the parton model), Chapter 9 (DGLAP evolution), and Chapter 12 (the synthesis). You can come back for the rest.

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Orientation: The Whole Story in Plain English

Before we dive into equations, let me sketch the entire story in ordinary words. If you forget every formula in this book but remember the story I am about to tell, you will still understand what is going on.

What we are trying to figure out

The question is simple: *what is a proton made of?* The proton is a ball of something roughly a femtometer (10^{-15} m) across, and we would like to know what is sloshing around inside it. We cannot cut it open, so we do the next best thing: we shoot electrons at it and watch what comes back out. Electrons are a particularly nice probe because we understand them completely — they are structureless, pointlike, and interact only through the electromagnetic force (and the weak force, which we will mostly ignore). Anything weird that happens in a scattering experiment is therefore due to the target, not the probe.

Three levels of seeing

If we shoot a gentle, low-energy electron at a proton, the electron barely notices anything internal — it sees a fuzzy blob of charge. The experiment measures a *form factor*, which is essentially the Fourier transform of the charge distribution. This is the non-relativistic, elastic regime, and it tells us the proton has a “size” of about 0.84 fm.

If we turn up the energy, the electron starts to resolve individual wiggles inside the blob. It can knock the proton into an excited state (a Δ^+ , an N^* , and so on), or break it apart into pions and other debris. This is the regime of excited baryons and the *inelastic* cross section.

Turn the energy up further still, and something magical happens: the electron starts scattering off what look like *pointlike* particles inside the proton. It is as if the fuzzy blob suddenly resolved into little hard marbles. Those marbles are the **partons**, and they are what Feynman proposed in 1969 as a way to explain the SLAC data. We now know the partons are quarks and gluons, and we know the theory that binds them: quantum chromodynamics, QCD.

Why the “pointlike marble” picture is so surprising

You would be justified in asking: if quarks are stuck inside the proton by an extremely strong force (strong enough that you cannot pull one out, ever), how can they possibly behave like free particles when the electron scatters off one? The answer is one of the deepest facts in all of particle

physics: the strong force *weakens* at short distances and *strengthens* at long distances. This is called **asymptotic freedom**. When a very energetic electron comes by, it probes the proton at such short distances that each quark inside it doesn't feel the others — for an instant, it's essentially free, and it behaves like a point charge bouncing off our electron. After the collision, the quarks pull themselves back together into hadrons (we never see a lone quark), but the hard scattering itself is clean and calculable.

The parton model in one sentence

Here is the whole naive parton model in a single sentence: *the cross section for ep scattering at high energies is what you'd get if you imagined the proton as a bag of free quarks, scattered the electron off one of them, and added up the contributions weighted by the probability that the struck quark was carrying a given fraction of the proton's momentum.*

That probability is called the **parton distribution function** $f_i(x)$: the probability density for finding a quark of type i carrying momentum fraction x . Parker, when you see $f_i(x)$, think: “how likely is it that, right now, there's a u -quark inside this proton carrying, say, 20% of the proton's total momentum?” That is literally what $f_u(0.2)$ answers.

Then QCD spoils the picture (gently)

If the quarks were really free, the measured structure function $F_2(x, Q^2)$ would depend only on x , not on Q^2 . This is called **Bjorken scaling**, and it was a spectacular SLAC discovery. But if you look more carefully, you find a small logarithmic dependence on Q^2 : at higher Q^2 , the function at large x goes *down* a bit, and the function at small x goes *up* a bit. Something is pushing probability from high x to low x as we look at the proton with a sharper resolution.

That something is *gluon emission*. A quark with momentum fraction x can radiate a gluon and become a quark with a smaller fraction $x' < x$. So as we turn up Q^2 , we resolve finer detail and more of the quarks look soft. The quantitative theory of how this happens is the **DGLAP equation**, named after Dokshitzer, Gribov, Lipatov, Altarelli, and Parisi. It tells us how $f_i(x, Q^2)$ evolves with Q^2 , and the evolution is driven by *splitting functions* $P_{ij}(z)$ that tell us the probability a parton of type j fluctuates into a parton of type i carrying fraction z of its momentum.

Fragmentation is the mirror image

When a quark gets knocked out of a proton, we never see the quark itself — we see a spray of hadrons (mostly pions). The **fragmentation function** $D_i^h(z)$ tells us the probability that a parton of type i hadronises into a specific hadron h carrying fraction z of its energy. PDFs tell us what's inside; fragmentation functions tell us what comes out. And, remarkably, they evolve with Q^2 according to the same DGLAP equations driven by the same splitting functions.

A glossary of intuitions

- **Cross section**: “how likely is it that this process happens.” Has units of area.
- **Form factor**: “how the target's internal structure suppresses what you'd see for a pointlike target.”

- **Structure function** (F_1 , F_2): “the inelastic generalisation of form factors; packages the proton’s internal response.”
- **Bjorken** x : “fraction of the proton’s momentum carried by the struck parton.” Between 0 and 1.
- Q^2 : “how hard we are whacking the proton”; more precisely, the (negative) virtuality of the exchanged photon.
- **Parton**: “quasi-free pointlike constituent of a hadron, seen in the infinite-momentum frame.”
- **PDF** $f_i(x, Q^2)$: “probability to find parton i carrying fraction x , when probing at resolution Q^2 .”
- **Asymptotic freedom**: “the strong coupling α_s gets *smaller* at higher energies.”
- **Scaling violations**: “the slow logarithmic dependence of F_2 on Q^2 — the PDFs evolve a bit as we change the resolution.”
- **Splitting function** $P_{ij}(z)$: “rate at which parton j splits into parton i with momentum fraction z .”
- **DGLAP**: “the differential equation that evolves PDFs in Q^2 .”
- **Fragmentation function** $D_i^h(z)$: “mirror-image PDF: rate at which parton i ends up as hadron h with fraction z .”

How to use the rest of this book

Part I goes through Griffiths §8.3–8.6 slowly: elastic scattering, the Rosenbluth formula, inelastic scattering, and the parton model. Part II follows Halzen & Martin §10.1–10.6: gluons enter the story, we meet the running coupling and the colour factors, and we derive the DGLAP equation. Part III (H&M §11.2) is fragmentation. Part IV is a synthesis chapter with a concrete road map for HW 5. Every chapter has worked examples in which I show every step of the derivation — not just the setup and the answer, but the algebra in between, so you can follow along even if you are uncomfortable with the more formal treatments in the textbooks.

Key idea

Every section in this book is designed so that you can read the boxed *Key idea* first and understand the core point in two sentences. If you then want to know where that core point comes from, read the derivation that precedes it. If you want to practise, do the *Self-check*. If you want to see the machinery in action, read the *Worked example*. You can traverse this book at whatever level of rigor you need for the day.

Part I

Electron–Proton Scattering and the Parton Model

Following Griffiths §8.3–8.6

Chapter 1

Elastic Electron–Proton Scattering

1.1 The point-target baseline: the Mott cross section

The simplest electron scattering experiment one can imagine is an electron fired at a static Coulomb field. In 1911, Rutherford computed the nonrelativistic cross section; for a relativistic, *point-like*, spin- $\frac{1}{2}$ target of infinite mass, the calculation was redone by Mott in 1929. In the lab frame, where the target is at rest and the incoming electron has energy E and scattering angle θ , the result is

$$\boxed{\left(\frac{d\sigma}{d\Omega}\right)_{\text{Mott}} = \frac{\alpha^2}{4E^2 \sin^4(\theta/2)} \cos^2(\theta/2)}, \quad (1.1)$$

in the high-energy limit in which the electron mass is negligible compared with E . Equation (1.1) has two pieces worth staring at:

- The $1/\sin^4(\theta/2)$ is the Rutherford singularity. It is hereditary: every amplitude mediated by the exchange of a massless gauge boson has a propagator $1/q^2$, and when you square and re-express q^2 in terms of the lab scattering angle you get $\sin^4(\theta/2)$. You can take this factor as the characteristic signature of single-photon exchange.
- The $\cos^2(\theta/2)$ factor is not hereditary; it is a consequence of the electron being a relativistic, spin- $\frac{1}{2}$ particle (in particular, of helicity conservation). It suppresses the backward cross section and says that at $\theta = \pi$ the cross section vanishes.

The Mott formula in (1.1) assumes an infinitely heavy target, so that there is no recoil: the outgoing electron energy E' equals the incoming E , and all the four-momentum transfer goes into the spatial components of q . Relaxing the infinite-mass assumption introduces a recoil correction. Let the target have mass M . Then for elastic scattering the outgoing energy is fixed by energy-momentum conservation to be

$$E' = \frac{E}{1 + (2E/M) \sin^2(\theta/2)}, \quad (1.2)$$

and the four-momentum transfer squared is

$$Q^2 \equiv -q^2 = 4EE' \sin^2(\theta/2) = 2M(E - E'). \quad (1.3)$$

The second equality uses the elastic constraint $(P + q)^2 = M^2$, which reduces to $2P \cdot q = Q^2$, or $2M\nu = Q^2$ in the lab frame with $\nu \equiv E - E'$. A compact way to encode the recoil is to write the cross section as Mott times E'/E :

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{Mott}}^{\text{recoil}} = \frac{\alpha^2}{4E^2 \sin^4(\theta/2)} \cos^2(\theta/2) \frac{E'}{E}. \quad (1.4)$$

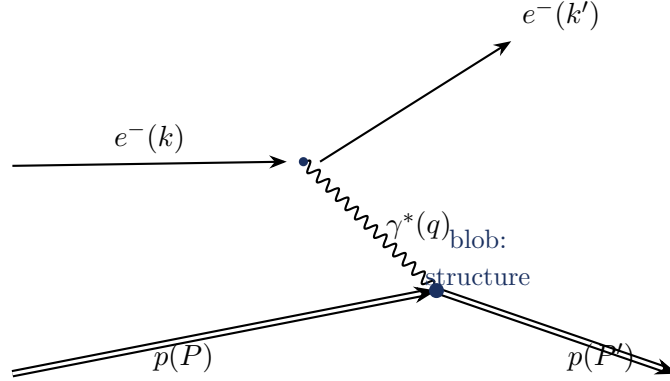


Figure 1.1: Elastic electron–proton scattering at the one-photon-exchange level. All of the proton’s internal complexity is hidden inside the hadronic vertex (the filled blob). Form factors parametrise exactly this complexity.

If you hear physicists say “Mott cross section” in the context of electron–proton scattering, this is usually what they mean.

Key idea

The Mott cross section is the benchmark against which all elastic lepton–hadron data are compared. Whenever a cross section is written as “ $(d\sigma/d\Omega)_{\text{Mott}} \times (\text{stuff})$,” the “stuff” encodes everything we do not yet know: form factors, structure functions, spin correlations.

1.2 The proton is not a point

If the proton were a pointlike Dirac particle of mass $M = 938 \text{ MeV}$ and charge $+e$, electron–proton scattering would be described by Eq. (1.4) plus a small magnetic-moment correction, and that would be that. When Hofstadter and his collaborators at Stanford began measuring elastic ep cross sections in the 1950s, what they found instead was that the measured $d\sigma/d\Omega$ fell off with increasing θ (hence with Q^2) much faster than the Mott prediction. The interpretation is physically immediate: the virtual photon has wavelength $\lambda \sim \hbar c/\sqrt{Q^2}$, and when this wavelength becomes comparable to the size of the proton (about 0.85 fm), the photon begins to “see” a spatially extended charge distribution rather than a single point. Scattering from different parts of the distribution interferes partially destructively and suppresses the cross section.

The simplest way to encode this smearing is by a *form factor*. If the charge is distributed according to a static density $\rho(\mathbf{x})$ satisfying $\int \rho(\mathbf{x}) d^3x = 1$, then the Born-approximation amplitude for scattering off the whole distribution is the point-target amplitude multiplied by

$$F(\mathbf{q}) = \int e^{i\mathbf{q}\cdot\mathbf{x}} \rho(\mathbf{x}) d^3x. \quad (1.5)$$

Thus $|F(\mathbf{q})|^2$ multiplies the Mott cross section, and $F(\mathbf{q} = 0) = 1$ recovers the total charge. The mapping $\rho \leftrightarrow F$ is a Fourier transform, so a proton of size $R \sim 1 \text{ fm}$ has a form factor that starts to turn over at $|\mathbf{q}| \sim 1/R \sim 0.2 \text{ GeV}$.

Margin note

Different assumed shapes for $\rho(\mathbf{x})$ give different form factors: a Gaussian gives a Gaussian, a hard sphere gives a sinc-like pattern, and an exponential $\rho \propto e^{-r/r_0}$ gives the *dipole* $F(Q^2) = (1 + Q^2 r_0^2)^{-2}$. The experimentally measured proton form factors are very well fit by a dipole with $r_0^{-2} \approx 0.71 \text{ GeV}^2$ over a wide range of Q^2 .

The static picture encoded in Eq. (1.5) is not quite adequate, however. It ignores the fact that the proton has a magnetic moment and that the electron's probe is relativistic. A proper treatment must use the covariant formalism of field theory, in which the electromagnetic current of the proton is not a scalar density but a four-vector operator $J_{\text{em}}^\mu(x)$. The full information about how the proton interacts with a photon is contained in the matrix element

$$\langle p(P') | J_{\text{em}}^\mu(0) | p(P) \rangle. \quad (1.6)$$

Lorentz covariance, parity invariance, and current conservation together force this matrix element to have a specific and highly constrained form.

1.3 The two proton form factors

Let $q = P' - P$ and $P_\mu = (P + P')_\mu/2$ (the average four-momentum). The matrix element in (1.6) must be a four-vector built bilinearly out of Dirac spinors $u(P)$, $\bar{u}(P')$, together with whatever four-vectors and gamma matrices are available. The candidates are γ^μ , P^μ , and q^μ . Current conservation $q_\mu J^\mu = 0$ kills any term proportional to q^μ (to leading order, one must be careful in general), and the Gordon identity

$$\bar{u}(P') \gamma^\mu u(P) = \bar{u}(P') \left[\frac{(P + P')^\mu}{2M} + \frac{i\sigma^{\mu\nu} q_\nu}{2M} \right] u(P) \quad (1.7)$$

lets us trade the P^μ -piece for one built from $\sigma^{\mu\nu}$, where $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$. The upshot is that *two* independent form factors suffice:

$$\langle p(P') | J_{\text{em}}^\mu | p(P) \rangle = e \bar{u}(P') \left[\gamma^\mu F_1(Q^2) + \frac{i\kappa \sigma^{\mu\nu} q_\nu}{2M} F_2(Q^2) \right] u(P) \quad (1.8)$$

Here $\kappa = 1.79$ is the anomalous magnetic moment of the proton (in nuclear magnetons), and $F_1(Q^2)$ and $F_2(Q^2)$ are the Dirac and Pauli form factors, respectively, normalised so that $F_1(0) = F_2(0) = 1$.

Key idea

For any spin- $\frac{1}{2}$ target, Lorentz and gauge symmetry force the electromagnetic vertex to be characterised by *exactly two* dimensionless functions of a single scalar, Q^2 . More structure (e.g. spin 3/2, as for the Δ) would allow more form factors; less structure (e.g. a Dirac pointlike) gives $F_1 \equiv 1$, $F_2 \equiv 0$.

A crucial physical remark is that at the single point $Q^2 = 0$, the two form factors take on specific physical meanings. $F_1(0) = 1$ gives the proton its total charge $+e$. The combination $F_1(0) + \kappa F_2(0) = 1 + 1.79 = 2.79$ gives the total proton magnetic moment in nuclear magnetons. At nonzero Q^2 the form factors encode the spatial extent of the charge and current distributions.

1.4 The Rosenbluth formula

To get an experimentally usable cross section from (1.8), one writes the spin-averaged hadronic tensor

$$L_{\text{hadr}}^{\mu\nu} = \frac{1}{2} \sum_{\text{spins}} \langle p(P') | J^\mu | p(P) \rangle^* \langle p(P') | J^\nu | p(P) \rangle, \quad (1.9)$$

contracts it with the leptonic tensor $L_{\mu\nu}^{\text{lept}}$ from QED, and divides by the square of the photon propagator $1/Q^4$. The algebra is tedious but mechanical, and the result is the *Rosenbluth formula* (M. Rosenbluth, 1950),

$$\left(\frac{d\sigma}{d\Omega} \right)_{ep \rightarrow ep} = \frac{\alpha^2}{4E^2 \sin^4(\theta/2)} \frac{E'}{E} \left\{ \left[F_1^2 - \frac{\kappa^2 Q^2}{4M^2} F_2^2 \right] \cos^2 \frac{\theta}{2} + \frac{Q^2}{2M^2} [F_1 + \kappa F_2]^2 \sin^2 \frac{\theta}{2} \right\}. \quad (1.10)$$

Notice the similarity to the Mott formula (1.4): there is the same $\alpha^2/4E^2 \sin^4(\theta/2)$ prefactor, the same E'/E recoil factor, and the same separation into a $\cos^2(\theta/2)$ piece (“electric”) and a $\sin^2(\theta/2)$ piece (“magnetic”). For a pointlike Dirac proton ($F_1 \equiv 1$, $F_2 \equiv 0$), Eq. (1.10) reduces to $(\alpha^2/4E^2 \sin^4(\theta/2))(E'/E)[\cos^2(\theta/2) + (Q^2/2M^2)\sin^2(\theta/2)]$, which is the Dirac formula.

1.4.1 Sachs form factors

It is computationally useful to define the Sachs electric and magnetic form factors,

$$G_E(Q^2) \equiv F_1(Q^2) - \tau \kappa F_2(Q^2), \quad (1.11)$$

$$G_M(Q^2) \equiv F_1(Q^2) + \kappa F_2(Q^2), \quad (1.12)$$

where $\tau \equiv Q^2/(4M^2)$ is a convenient dimensionless variable. In terms of the Sachs functions, the Rosenbluth formula takes a manifestly positive form:

$$\left(\frac{d\sigma}{d\Omega} \right)_{ep} = \frac{\alpha^2 \cos^2(\theta/2)}{4E^2 \sin^4(\theta/2)} \frac{E'}{E} \frac{1}{1 + \tau} \left[G_E^2 + \frac{\tau}{\varepsilon} G_M^2 \right], \quad (1.13)$$

where $\varepsilon = [1 + 2(1 + \tau) \tan^2(\theta/2)]^{-1}$ is the virtual photon polarisation parameter. The two Sachs form factors are normalised by $G_E(0) = 1$ (charge) and $G_M(0) = 1 + \kappa = 2.79$ (magnetic moment). Experimentally, both are well described at small and moderate Q^2 by the dipole form

$$G_E(Q^2) \approx \frac{G_M(Q^2)}{1 + \kappa} \approx G_D(Q^2) \equiv \frac{1}{(1 + Q^2/0.71 \text{ GeV}^2)^2}, \quad (1.14)$$

though more precise experiments have now revealed small but interesting deviations from this simple parameterisation at large Q^2 .

1.4.2 Rosenbluth separation

How does one extract G_E and G_M from experimental cross sections? Rearrange Eq. (1.13):

$$\underbrace{\frac{(d\sigma/d\Omega)(1 + \tau)}{(d\sigma/d\Omega)_{\text{Mott}}}}_{\text{LHS}} = G_E^2(Q^2) + \frac{\tau}{\varepsilon} G_M^2(Q^2). \quad (1.15)$$

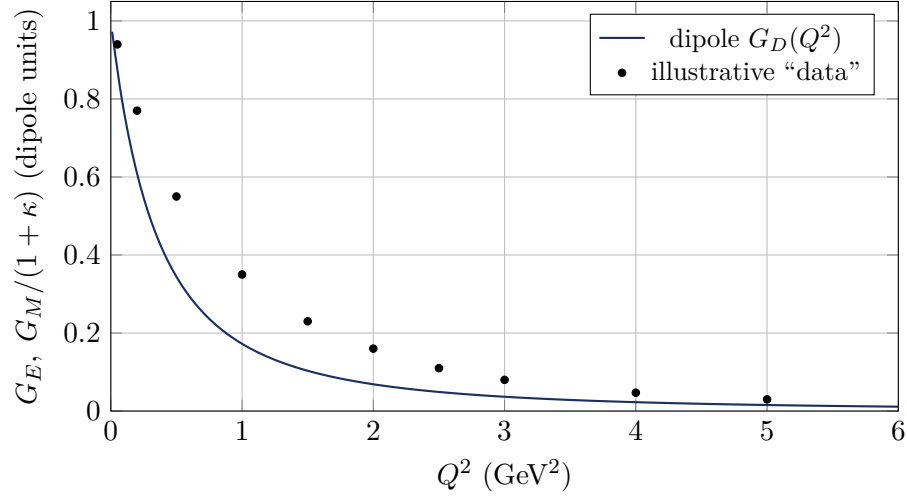


Figure 1.2: Schematic dipole form factor versus Q^2 , with illustrative points simulating real Rosenbluth-separated data. The form factor falls rapidly: at $Q^2 = 1 \text{ GeV}^2$ the cross section is suppressed by a factor of roughly $G_D^2 \sim 0.12$ relative to a pointlike proton.

At *fixed* Q^2 , we can vary ε (by varying the beam energy E and the scattering angle θ in tandem, keeping Q^2 constant), and the LHS becomes a linear function of $1/\varepsilon$. Plotting the experimentally measured LHS against $1/\varepsilon$ yields a straight line whose slope is τG_M^2 and whose intercept is G_E^2 . This is the *Rosenbluth separation*. Repeating the procedure at many values of Q^2 gives the two form factors as functions of Q^2 .

Margin note

In the modern era, the Rosenbluth method has been partially superseded by *polarisation transfer* experiments, which measure the ratio G_E/G_M directly from the polarisation of the recoiling proton. These experiments revealed a startling deviation from the “dipole equality” at large Q^2 : the ratio $\mu_p G_E/G_M$ falls almost linearly with Q^2 , reaching values as small as 0.2 around $Q^2 \sim 8 \text{ GeV}^2$. This is an active area of research as of 2024.

1.5 From form factors to charge distributions

Inverting a Fourier transform, one can interpret G_E as the Fourier transform of an effective charge distribution in the proton’s Breit frame (the frame in which q^μ has no time component). At small Q^2 , expanding the form factor gives

$$G_E(Q^2) \approx 1 - \frac{1}{6} \langle r^2 \rangle Q^2 + \dots \quad (1.16)$$

from which one reads off the proton’s root-mean-square charge radius,

$$\sqrt{\langle r^2 \rangle_E} \approx 0.84\text{--}0.88 \text{ fm}, \quad (1.17)$$

depending on the precise experimental method. (This quantity was the subject of the celebrated “proton radius puzzle” that began in 2010, when muonic hydrogen spectroscopy gave a value significantly smaller than electron-scattering measurements; the discrepancy has been largely

resolved.) The magnetic radius is similar in magnitude. In any case, the proton is decisively *not* a point on any scale probed by GeV electrons.

1.6 Where elastic scattering runs out

Now for the key realisation. Elastic scattering requires the proton to survive the collision unchanged. This requires that the invariant mass of the hadronic final state be exactly M , or equivalently, that $Q^2 = 2M\nu$ (the “elastic line” in the (Q^2, ν) -plane). But as we increase the beam energy, and as the virtual photon transfers more and more energy ν to the proton, the proton can also be *excited* into one of its resonance states (Δ , N^* , ...), or broken up entirely into a shower of mesons. In a word, the scattering becomes *inelastic*. The mathematical implication is clear: the final-state invariant mass W is now a second independent variable, and the cross section depends on two scalars instead of one.

This is where Griffiths § 8.3 ends and § 8.4 begins. It is also where the physics gets much richer.

Self-check

(1) Show, directly from four-momentum conservation, the recoil formula (1.2). (2) Verify that the Rosenbluth formula (1.10) reduces to the Dirac cross section when $F_1 = 1$, $F_2 = 0$. (3) Expand G_E to first nontrivial order in Q^2 and confirm the coefficient of the charge-radius term. (4) What is the *magnetic* charge radius of the proton, in terms of the dipole parameter $r_0^2 = 12/0.71 \text{ GeV}^{-2}$?

Chapter 2

Inelastic Electron–Proton Scattering

2.1 Opening up the hadronic final state

Above a few hundred MeV of energy loss ν , the virtual photon has enough room to do more than just bounce elastically off the proton. It can excite the proton into a Δ^+ (mass 1232 MeV), or into an N^* (Roper, 1440 MeV, and higher), or into a two-particle final state like $p\pi^0$, $n\pi^+$; at still larger ν it can produce three-, four-, and many-hadron final states. The distribution of final-state invariant masses $W^2 = (P + q)^2$ shows a set of increasingly overlapping resonance peaks and then, in the “deep” regime $W^2 \gtrsim 4 \text{ GeV}^2$, a smooth continuum.

Rather than attempting to resolve each exclusive channel, experimenters made a virtue out of necessity by measuring the *inclusive* cross section for

$$e^-(k) + p(P) \longrightarrow e^-(k') + X(P_X), \quad (2.1)$$

where X is *any* hadronic final state. One only sees the outgoing electron (and its energy E' and angle θ); one sums over everything else. This is all the electromagnetic probe can possibly tell you about the proton, and as we shall see, it turns out to tell you a great deal.

2.2 A dictionary of kinematic variables

Inclusive DIS has two independent Lorentz scalars. There are many popular choices for which two. Here is a dictionary:

$$q^\mu \equiv k^\mu - k'^\mu, \quad Q^2 \equiv -q^2 = 4EE' \sin^2(\theta/2), \quad (2.2)$$

$$\nu \equiv \frac{P \cdot q}{M} = E - E' \text{ (lab)}, \quad W^2 \equiv (P + q)^2 = M^2 + 2M\nu - Q^2, \quad (2.3)$$

$$x \equiv \frac{Q^2}{2P \cdot q} = \frac{Q^2}{2M\nu}, \quad y \equiv \frac{P \cdot q}{P \cdot k} = \frac{\nu}{E} \text{ (lab)}. \quad (2.4)$$

The pair (Q^2, ν) is natural for the lab frame. The pair (x, Q^2) is natural for the parton model, because we will see that x is the fraction of the proton’s momentum carried by the struck parton. The pair (x, y) is natural for charged-current neutrino scattering. The pair (W^2, Q^2) is natural for resonance physics. Any two of these variables can be chosen as independent; the rest are determined.

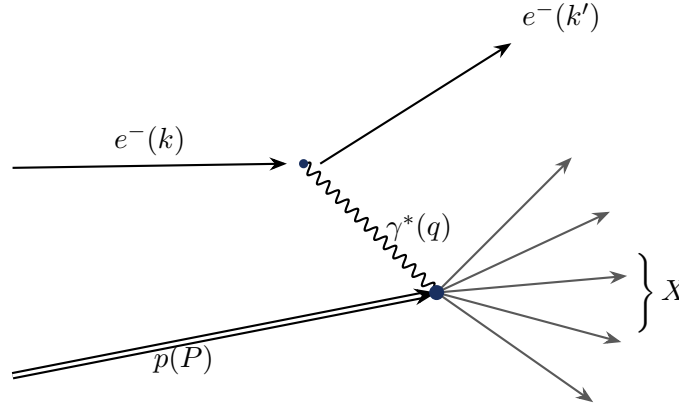


Figure 2.1: Inclusive deep inelastic scattering. The hadronic final state X is unresolved and summed over; the blob encodes all the possible ways a proton of momentum P can absorb a virtual photon of momentum q . All the nontrivial QCD lives inside the blob.

Key idea

Physical meaning in one breath. Q^2 measures how hard the photon kicks; ν measures how much energy it delivers; W is the mass of the resulting hadronic debris; x measures the “pointlikeness” of the target constituent struck; y measures the fractional energy lost by the electron.

The elastic case $W = M$ corresponds to $Q^2 = 2M\nu$, i.e. $x = 1$. Thus $x = 1$ is the elastic limit, and the interior of the x -interval $(0, 1)$ is where inelastic events live. $x > 1$ is kinematically forbidden (the hadronic final state would have to have $W^2 < M^2$), and $x = 0$ corresponds to $Q^2 = 0$ at fixed ν —the real-photon limit.

2.3 The inelastic structure functions $W_{1,2}$

We now want to repeat, for the inclusive cross section, the analysis that led to the Rosenbluth formula. The leptonic side is unchanged—it is still QED, and a single photon is exchanged. The relevant matrix element on the hadronic side is now

$$W^{\mu\nu}(P, q) \equiv \frac{1}{4\pi M} \sum_X (2\pi)^4 \delta^{(4)}(P + q - P_X) \langle p(P) | J^{\mu\dagger} | X \rangle \langle X | J^\nu | p(P) \rangle, \quad (2.5)$$

which is the inclusive analog of the elastic matrix element, summed over all hadronic final states X . The tensor $W^{\mu\nu}$ is the *hadronic tensor*. It is real, symmetric, and (by current conservation) transverse: $q_\mu W^{\mu\nu} = q_\nu W^{\mu\nu} = 0$. The most general tensor built out of P , q , and $g^{\mu\nu}$, subject to parity and transversality, contains two independent scalar functions called the *structure functions*:

$$W^{\mu\nu}(P, q) = \left(-g^{\mu\nu} + \frac{q^\mu q^\nu}{q^2} \right) W_1(\nu, Q^2) + \left(P^\mu - \frac{P \cdot q}{q^2} q^\mu \right) \left(P^\nu - \frac{P \cdot q}{q^2} q^\nu \right) \frac{W_2(\nu, Q^2)}{M^2}. \quad (2.6)$$

Here W_1 and W_2 are real, dimensional, Lorentz-invariant functions of the two independent scalars ν and Q^2 . They contain *all* the information about how the proton responds to an electromagnetic probe.

2.3.1 From hadronic tensor to cross section

Contracting $W^{\mu\nu}$ with the QED leptonic tensor

$$L_{\text{lept}}^{\mu\nu} = 2(k^\mu k'^\nu + k'^\mu k^\nu - g^{\mu\nu} k \cdot k'), \quad (2.7)$$

dividing by Q^4 (the photon propagator squared), and re-expressing the result in lab-frame variables, one finds the inclusive double differential cross section:

$$\boxed{\frac{d^2\sigma}{dE' d\Omega} = \frac{\alpha^2}{4E^2 \sin^4(\theta/2)} \left[W_2(\nu, Q^2) \cos^2(\theta/2) + 2W_1(\nu, Q^2) \sin^2(\theta/2) \right]}. \quad (2.8)$$

This is the inelastic analog of the Rosenbluth formula. The W_1 piece, multiplied by $\sin^2(\theta/2)$, is the “magnetic” response (transverse photon polarisation); the W_2 piece, multiplied by $\cos^2(\theta/2)$, is the “electric” response (a mixture of longitudinal and transverse).

Margin note

In the elastic limit, the structure functions become delta functions supported on the line $W = M$:

$$W_1^{\text{el}} = \frac{Q^2}{4M^2} G_M^2(Q^2) \delta(\nu - \frac{Q^2}{2M}), \quad W_2^{\text{el}} = \frac{G_E^2 + \tau G_M^2}{1 + \tau} \delta(\nu - \frac{Q^2}{2M}).$$

Substituting into Eq. (2.8) and integrating the delta function reproduces the Rosenbluth formula, as a consistency check.

2.3.2 What experimenters expected, and what they saw

In the mid-1960s, when Friedman, Kendall, Taylor and the MIT–SLAC group began mapping out $W_1(\nu, Q^2)$ and $W_2(\nu, Q^2)$ up to $Q^2 \sim 8 \text{ GeV}^2$, the community expected the inelastic structure functions to fall off rapidly with Q^2 , just as the elastic form factors do. After all, “bigger means softer” was the whole lesson of the Hofstadter era.

What they actually found was the opposite. At *fixed* $\omega \equiv 1/x = 2M\nu/Q^2$, the measured νW_2 was essentially *independent* of Q^2 over a factor of 10 or more in Q^2 . The virtual photon appeared to be scattering off something *pointlike* inside the proton—something with no structure of its own on the scale probed, so that the cross section refused to fall with Q^2 . This astonishing experimental result is *Bjorken scaling*, first recognised by James Bjorken in 1968 on the basis of current-algebra sum rules, and immediately interpreted by Richard Feynman in terms of his soon-to-be-famous parton model. It earned Friedman, Kendall and Taylor the 1990 Nobel Prize.

2.4 Statement of Bjorken scaling

Precisely, the statement of Bjorken scaling is:

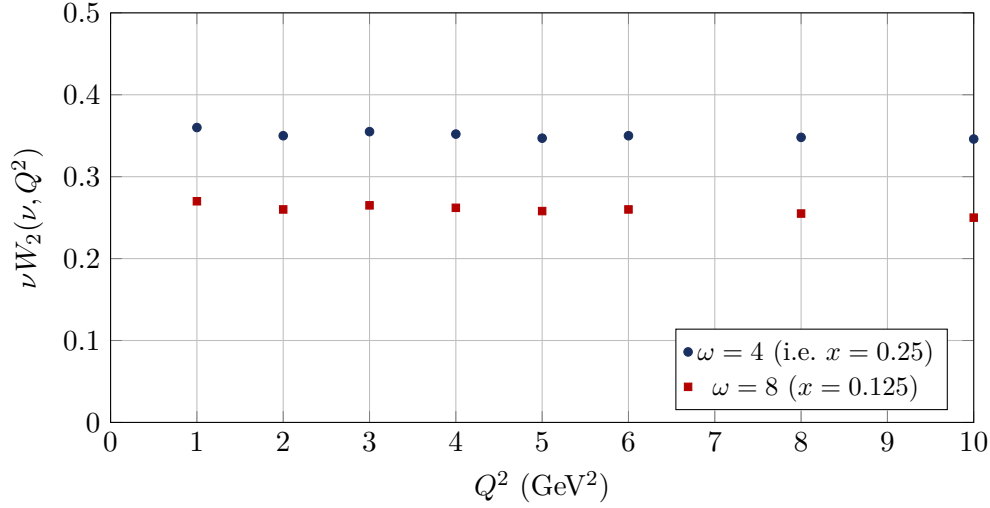


Figure 2.2: Schematic depiction of Bjorken scaling. At two fixed values of $\omega = 1/x$, the measured νW_2 is essentially independent of Q^2 . Had the proton been a finite-sized lump of charge described by a dipole form factor, the data points would have fallen by almost two orders of magnitude across the displayed Q^2 range.

Bjorken scaling. In the limit $Q^2 \rightarrow \infty$, $\nu \rightarrow \infty$ with $x = Q^2/(2M\nu)$ fixed, the structure functions become functions of x alone:

$$M W_1(\nu, Q^2) \rightarrow F_1(x), \quad \nu W_2(\nu, Q^2) \rightarrow F_2(x). \quad (2.9)$$

The dimensionless functions $F_1(x)$ and $F_2(x)$ are what we will hereafter call the “structure functions” without further qualification; they depend on a single dimensionless variable, and $(2MW_1, \nu W_2) \rightarrow (2xF_1, F_2)$ is more or less the universal notation in the modern literature.

That the structure functions have scaled to functions of a pure number is a strong hint that there is a dimensionless physical process underneath—one whose dynamics are not set by any hadronic mass scale such as M_p or Λ_{QCD} . Something must be pointlike. The next chapter is devoted to understanding what that something is.

Self-check

- (1) Starting from $q_\mu W^{\mu\nu} = 0$, derive the Lorentz structure in Eq. (2.6).
- (2) Show that in the elastic limit the scaling functions $F_{1,2}(x)$ become delta functions of $x - 1$, and identify the coefficients in terms of G_E and G_M .
- (3) Express the differential cross section (2.8) in terms of the variables (x, Q^2) and show that the Jacobian is $\pi/(EE'/x)$ or equivalently $d\nu dQ^2 = (2ME/(y)) dx dy$.
- (4) What would Bjorken scaling look like if the virtual photon were scattering off a Gaussian blob of charge of radius 0.5 fm? Sketch F_2 vs. Q^2 at fixed x .

Chapter 3

The Parton Model

3.1 Feynman’s picture

Richard Feynman’s proposal for what Bjorken scaling meant physically is both radical and childishly simple. He imagined the proton not as a continuous distribution of charge, not even as a bound state of three constituent quarks, but as a collection of free, on-shell, pointlike objects—the *partons*—moving *collinearly* with the proton’s three-momentum at very high energy. A given parton i carries a fraction ξ_i of the proton’s four-momentum, with $\sum_i \xi_i = 1$. The partons do not interact with each other during the short time $\tau_{\text{int}} \sim 1/Q$ that the virtual photon is near the proton; they only interact on the (much longer) hadronic timescale $\tau_{\text{had}} \sim 1/\Lambda_{\text{QCD}}$.

This is best visualised in the *infinite-momentum frame*, a Lorentz frame in which the proton is moving at nearly the speed of light. In such a frame, time dilation stretches τ_{had} by the Lorentz factor $\gamma = E_{\text{proton}}/M$, while the interaction time τ_{int} is a Lorentz invariant. For γ large enough, the two timescales separate cleanly:

$$\tau_{\text{int}} \ll \tau_{\text{had}}^{(\text{lab})} \ll \tau_{\text{had}}^{(\text{IMF})}. \quad (3.1)$$

During the brief window τ_{int} that the photon interacts with the proton, the partons’ own interactions are effectively frozen. The photon picks one parton, elastically scatters off it as if it were free, and flies away. The hadronic fragmentation of the struck parton (and of the remnant) occurs much later, by which time the lepton has long since recorded its scattering angle and left the interaction region.

Key idea

The parton model is a *separation of timescales*. If the hard interaction is fast enough, you can treat the constituents as free, even though in reality they are permanently confined. Everything in Part I of this document and nearly everything in Part II is an elaboration of this one idea.

3.2 On-shellness of the struck parton fixes $\xi = x$

Consider a parton of momentum ξP (collinear with the proton). The virtual photon transfers momentum q and turns this parton into a final-state parton of momentum $\xi P + q$. Assuming the final parton is (approximately) massless, we require

$$(\xi P + q)^2 = 0. \quad (3.2)$$

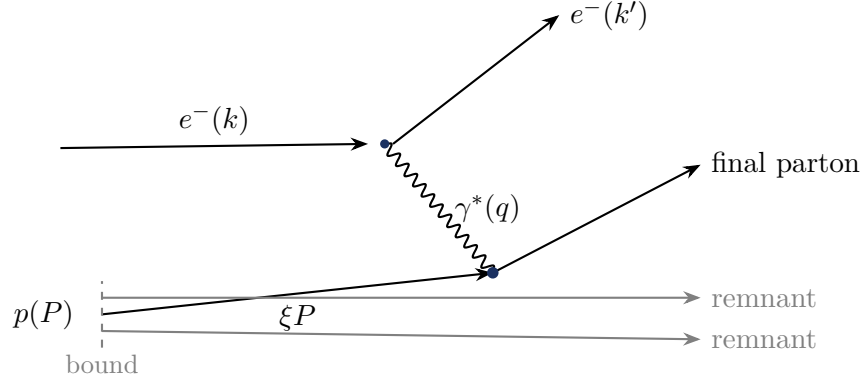


Figure 3.1: Parton-level picture of deep inelastic scattering. The proton is modelled as a collinear beam of free partons; the virtual photon picks one, scatters off it elastically, and the remnant flies on unaffected. The momentum fraction ξ of the struck parton is forced by on-shell kinematics to equal $x = Q^2/(2M\nu)$.

Expanding gives $\xi^2 M^2 + 2\xi P \cdot q - Q^2 = 0$. In the deep inelastic limit $Q^2 \gg M^2$ the first term is negligible, and we find $\xi = Q^2/(2P \cdot q) = x$. *The Bjorken variable x is exactly the momentum fraction of the parton that was struck.*

This is the single most important kinematic identity in deep inelastic scattering. The abstract ratio $x = Q^2/(2M\nu)$, which had until now been just a convenient dimensionless combination, has acquired an immediate physical meaning.

3.3 Parton distribution functions

Let $f_i(\xi)$ denote the probability density for finding a parton of type i carrying a fraction ξ of the proton's momentum. More precisely, $f_i(\xi) d\xi$ is the probability that a randomly chosen parton of type i has $\xi_i \in (\xi, \xi + d\xi)$. The functions $f_i(\xi)$ are the *parton distribution functions* (PDFs). In the parton model they are intrinsic properties of the proton, measured (not computed) from data. They are nonnegative, they satisfy the normalisation conditions to be discussed below, and they vanish at $\xi = 0$ and $\xi = 1$.

The inclusive ep cross section in the parton model is an incoherent sum over parton species and an integral over momentum fractions, with a single-parton cross section $d\hat{\sigma}_i$ inside:

$$d\sigma = \sum_i \int_0^1 d\xi f_i(\xi) d\hat{\sigma}_i(\xi P, k, k'). \quad (3.3)$$

The partonic cross section $d\hat{\sigma}_i$ is, by assumption, simply the Born QED cross section for scattering a point Dirac particle of charge e_i off a point electron. A short calculation (see also Chapter 6) gives

$$\frac{d\hat{\sigma}_i}{dQ^2} = \frac{2\pi\alpha^2 e_i^2}{Q^4} [1 + (1 - \hat{y})^2] \delta\left(\xi - \frac{Q^2}{2\xi P \cdot q}\right), \quad (3.4)$$

where $\hat{y} = (\xi P \cdot q)/(\xi P \cdot k) = y$ is the parton-level inelasticity. The δ enforces on-shellness of the final parton, $\xi = x$. Doing the ξ integral collapses the sum to

$$\frac{d^2\sigma}{dx dy} = \frac{4\pi\alpha^2 s}{Q^4} \left[1 - y + \frac{y^2}{2}\right] \sum_i e_i^2 x f_i(x), \quad (3.5)$$

from which we extract the scaling structure functions

$$F_2(x) = \sum_i e_i^2 x f_i(x), \quad 2xF_1(x) = F_2(x). \quad (3.6)$$

The first equation is the *parton-model formula* for the structure function F_2 : it is a weighted sum of parton distributions, with the weights equal to the squared electric charges of the partons. The second is the *Callan–Gross relation*: it says that the longitudinal structure function $F_L \equiv F_2 - 2xF_1$ vanishes identically. This is a non-trivial consequence of scattering off *spin- $\frac{1}{2}$* partons: for spin-0 partons one would find instead $2xF_1 = 0$, and for a magnetic interaction one would find something else. The Callan–Gross relation was tested experimentally almost immediately after the parton model was proposed, and its confirmation was an important clue that the partons were spin- $\frac{1}{2}$ (consistent with their identification as quarks).

3.4 Scaling is automatic

Notice that the right-hand sides of Eq. (3.6) are manifestly *independent* of Q^2 . The parton distribution functions $f_i(x)$ depend on x alone. Therefore F_2 depends on x alone. Therefore, Bjorken scaling holds as an exact property of the parton model. The astonishing experimental finding of the late 1960s is, in this language, trivially explained: it is a kinematic consequence of photons scattering elastically off free pointlike constituents.

Of course scaling is only approximately true in nature, because the partons in real QCD are not truly free; they interact weakly, not trivially, and the deviations from exact scaling carry information about the strength of those interactions. But that is the subject of Part II.

Margin note

Parton model vs. quark model vs. QCD. Feynman’s “partons” were agnostic about their identity—they could have been quarks, or pions, or anything else pointlike. The identification partons = quarks came later, from the Callan–Gross relation (spin $\frac{1}{2}$), the ratio $\sigma(e^+e^- \rightarrow \text{hadrons})/\sigma(e^+e^- \rightarrow \mu^+\mu^-)$ (fractional charges), and sum rules on the parton distributions (flavour quantum numbers). The identification quarks = constituents of a color-SU(3) gauge theory (i.e., QCD) came even later. One should keep these three steps distinct mentally.

3.5 Inclusive vs. exclusive, coherent vs. incoherent

Why does the parton model sum cross sections *incoherently*? Why not add amplitudes?

The answer, in a phrase, is “quantum mechanics of a fast moving bound state.” The partons inside the proton have a wide distribution of longitudinal momenta. When the virtual photon interacts with a parton of momentum xP , the other partons are spectators—they do not participate in the interaction, and their quantum state is unchanged. After the interaction, the struck parton is off-shell (well, actually, it is on-shell; its momentum has been shifted by q so that it satisfies $(xP + q)^2 = 0$), but the *other* partons are now in a different quantum state from the initial proton, because the proton has lost the one parton that was struck. In an exclusive final-state measurement (where we try to identify the individual hadrons), this mismatch matters: amplitudes for different partons do add up coherently into a more complicated object. In the *inclusive* measurement we are making—where we sum over all hadronic final states X —this coherence is lost; the sum-over- X

turns the amplitude squared into a probability sum, and one recovers the incoherent parton-model formula.

This argument can be made precise using the *optical theorem* for forward virtual Compton scattering: $W^{\mu\nu} \propto \text{Im } T_{\text{fwd}}^{\mu\nu}$, and the forward-Compton amplitude is what is computed parton by parton with single-particle cuts of the forward diagram.

3.6 A first look at the partons

Even before saying anything about QCD, one can learn a lot from the parton model:

1. *The partons that couple electromagnetically are fractionally charged.* The ratio $R \equiv \sigma(e^+e^- \rightarrow \text{hadrons})/\sigma(e^+e^- \rightarrow \mu^+\mu^-)$ equals $N_c \sum_q e_q^2$ in the parton model, where N_c is the number of colors and the sum runs over energetically accessible quark flavors. Below charm threshold one finds $R = 3(\frac{4}{9} + \frac{1}{9} + \frac{1}{9}) = 2$; between c and b thresholds, $R = 3(\frac{4}{9} + \frac{1}{9} + \frac{1}{9} + \frac{4}{9}) = 10/3$; above b , $R = 11/3$. Experimentally confirmed.
2. *The partons are spin- $\frac{1}{2}$.* Callan–Gross.
3. *There is a factor of $N_c = 3$.* The absolute rate of $e^+e^- \rightarrow \text{hadrons}$ requires $N_c = 3$. This is the first direct experimental evidence for color. Without color, the prediction would be a factor of 3 too small.

From DIS alone, without yet knowing anything about how partons become hadrons, one can map out the sum $\sum_i e_i^2 x f_i(x)$. From e^+e^- , one can determine $\sum_q e_q^2$. From neutrino DIS (see Griffiths §9 or H&M §12), one can separately determine $x f_q(x) + x f_{\bar{q}}(x)$ and $x f_q(x) - x f_{\bar{q}}(x)$. Combining all three kinds of data, one can extract the individual flavor PDFs. That is what we turn to next.

Self-check

- (1) Show from Eq. (3.3) that $2xF_1 = F_2$ for scattering off spin- $\frac{1}{2}$ partons. Where would the argument break for spin-0 partons? (2) Compute R above the $b\bar{b}$ threshold, assuming three colors and the six flavors u, d, s, c, b . (3) Why does the parton-model derivation require the infinite-momentum frame? What would go wrong if one tried to do it in the proton rest frame?

Chapter 4

Quark Distribution Functions

4.1 Valence and sea

In the strict static quark model, the proton is a bound state of exactly three quarks: two u 's and a d . If that were the whole story, the parton distributions would be simple: $u(\xi)$ and $d(\xi)$ would each be sharply peaked at $\xi = 1/3$, and $\bar{u}(\xi) = \bar{d}(\xi) = s(\xi) = \dots = 0$. The actual parton distributions are very much not like that. The two reasons are:

1. The three valence quarks are not independent; they exchange gluons, which redistributes their momentum over a broad range of ξ .
2. The gluons can (and do) split into $q\bar{q}$ pairs of any flavor. The proton at any instant therefore contains not only its three valence quarks but also a *sea* of virtual $q\bar{q}$ pairs (for every flavor, including heavy quarks) plus the gluons that glue it all together.

We therefore split every flavor's distribution into a valence and a sea piece,

$$u(\xi) = u_V(\xi) + u_S(\xi), \tag{4.1}$$

$$d(\xi) = d_V(\xi) + d_S(\xi), \tag{4.2}$$

$$\bar{u}(\xi) = u_S(\xi), \quad \bar{d}(\xi) = d_S(\xi), \tag{4.3}$$

$$s(\xi) = \bar{s}(\xi), \quad c(\xi) = \bar{c}(\xi), \quad \dots \tag{4.4}$$

The sea distributions are taken to be charge-symmetric in the lightest flavors to a first approximation. In reality, $\bar{d}(\xi) \neq \bar{u}(\xi)$ (the “Gottfried sum rule violation”) and the strange sea is different from the nonstrange sea; these are small effects that we won't dwell on.

4.2 Flavor sum rules

The proton has baryon number +1, electric charge +1, and zero strangeness. In terms of PDFs this means, for any Q^2 ,

$$\int_0^1 d\xi [u(\xi) - \bar{u}(\xi)] = 2, \quad (4.5)$$

$$\int_0^1 d\xi [d(\xi) - \bar{d}(\xi)] = 1, \quad (4.6)$$

$$\int_0^1 d\xi [s(\xi) - \bar{s}(\xi)] = 0, \quad (4.7)$$

$$\int_0^1 d\xi [c(\xi) - \bar{c}(\xi)] = 0, \dots \quad (4.8)$$

These are exact consequences of conservation laws, and they hold at every Q^2 . In the valence/sea decomposition they become

$$\int_0^1 u_V(\xi) d\xi = 2, \quad \int_0^1 d_V(\xi) d\xi = 1, \quad (4.9)$$

which merely counts the number of valence quarks of each flavor.

A much more striking sum rule involves the first *moment*:

$$\int_0^1 d\xi \xi \left[\sum_q f_q(\xi) + \sum_{\bar{q}} f_{\bar{q}}(\xi) \right] \approx 0.5. \quad (4.10)$$

The left-hand side is the *momentum fraction carried by charged partons*. One might naïvely expect it to be unity—“everything the proton is made of should carry all its momentum”—but experimentally it is only about half. The missing half of the proton’s momentum is carried by partons that do not couple to the electromagnetic probe: electrically neutral particles, which is to say, gluons.

Key idea

Half the proton’s momentum is invisible to photons. This was the first experimental hint that gluons not only exist but are *major* constituents of the proton—the most direct early evidence for them, predating the discovery of three-jet events in e^+e^- annihilation.

4.3 Shapes of the distributions

It is worth committing to memory a qualitative picture of the parton distributions at a moderate resolution (say, $Q^2 \sim 10 \text{ GeV}^2$), because all of QCD phenomenology at colliders rests on this picture.

The features to remember from Fig. 4.1 are:

- **Valence peaks.** u_V and d_V peak between $x \approx 0.1$ and $x \approx 0.3$, with $u_V \approx 2d_V$ over most of that range (from the flavor sum rules). Both vanish at $x = 0$ (no valence quarks at zero momentum fraction) and at $x = 1$ (no single quark carries the whole proton’s momentum).
- **Sea rises at small x .** The antiquark PDFs rise like $1/x^\alpha$ with α a little greater than $\frac{1}{2}$, because gluon-to-sea splitting gives a soft singularity. At $x \lesssim 0.05$, the sea dominates over the valence.

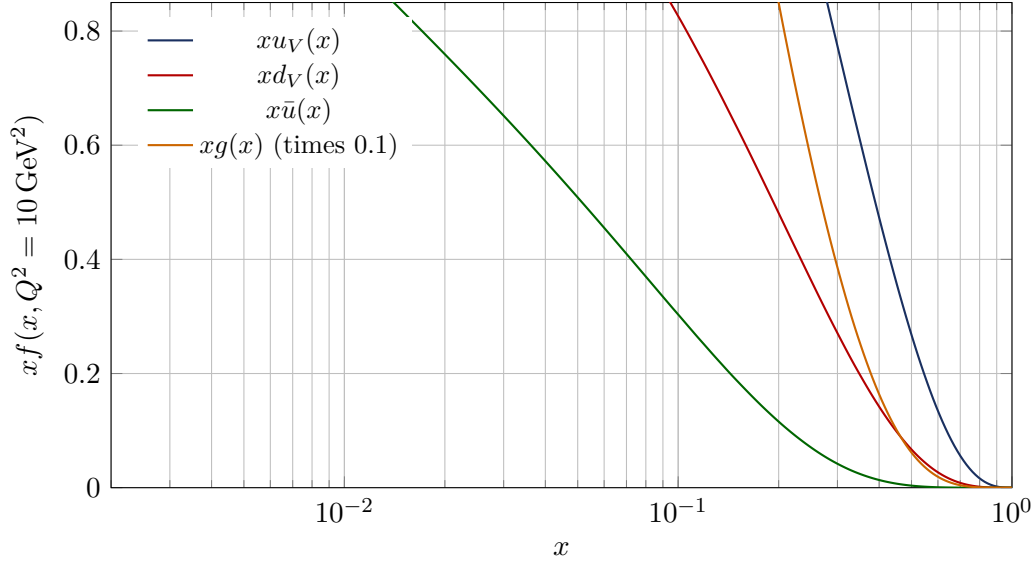


Figure 4.1: Schematic parton distributions in the proton at $Q^2 = 10 \text{ GeV}^2$. The valence u and d (blue and red) peak near $x \approx 0.15$ with u_V about twice d_V , as required by the flavor sum rules. The sea (green) rises rapidly at small x . The gluon (orange, shown multiplied by 0.1 for display) dominates below $x \approx 0.1$. Exact shapes are illustrative; real PDFs are extracted from global fits.

- **Gluon dominates at low x .** The gluon PDF rises even faster and is the dominant component below $x \approx 0.1$. At $x \sim 10^{-3}$ and $Q^2 \sim 10 \text{ GeV}^2$, $xg(x)$ is on the order of 10—so xg looks “tall” in such plots, and a factor of 0.1 is often applied for display purposes.
- **Nothing much above $x \gtrsim 0.7$.** Phase space is exhausted; PDFs fall like $(1-x)^n$ with $n \approx 3$ for u_V , $n \approx 4$ for d_V , and $n \approx 5$ for sea and gluons (“counting rules” attributed to Brodsky and Farrar).

4.4 A quick sanity check: the ratio F_2^N/F_2^P

A nice check of the parton picture comes from comparing deep inelastic scattering off protons and neutrons. Neglecting the sea and the strange quark for simplicity, and using isospin symmetry (so that $u^n(\xi) = d^p(\xi)$ and $d^n(\xi) = u^p(\xi)$), one finds

$$\frac{F_2^N(x)}{F_2^P(x)} = \frac{u^p(x) + 4d^p(x)}{4u^p(x) + d^p(x)}. \quad (4.11)$$

For $u = 2d$ (*naïve* valence counting), this ratio is $(2 + 4)/(8 + 1) = 2/3$. As $x \rightarrow 1$, if $d/u \rightarrow 0$ (“ u -quark dominance at the endpoint”), the ratio approaches $1/4$. The experimental data—from measurements on deuterium and separately on free protons—confirm this behavior qualitatively, with the ratio falling from ~ 1 at small x to $\sim 1/4$ at large x .

4.5 Where do the distributions come from?

Parton distribution functions are intrinsically nonperturbative: they encode long-distance bound-state dynamics of QCD that we cannot compute from first principles with any current technology.

Lattice QCD can compute their low moments directly, and in recent years “quasi-PDF” and “pseudo-PDF” techniques have made significant progress toward computing the full x -dependence. For phenomenology, however, PDFs are always extracted from global fits to experimental data—from DIS, from Drell–Yan, from jet production, from W/Z production, and more recently from LHC data at the TeV scale. Groups like CT (CTEQ), MSHT (MSTW), NNPDF, ABMP, and HERAPDF publish regular updates. A modern PDF set typically contains 20–50 “error sets” that quantify the uncertainty from the data fit.

What QCD *does* predict, and predict very precisely, is how the PDFs change as you vary the resolution Q^2 at which you probe them. This is the subject of Part II.

Self-check

(1) From the sum rules (4.5)–(4.6) and the ratio $u_V = 2d_V$ at the *integral* level, what can you conclude about the pointwise equality $u_V(x) = 2d_V(x)$? Is it required? (2) The momentum sum rule including gluons is $\int x[\sum_q(q + \bar{q}) + g] dx = 1$ exactly. If the charged momentum fraction is 0.5, how much momentum is in gluons? (3) Show that in the limit where only the u and d valence quarks contribute, F_2^p/F_2^n is bounded between 1/4 and 4.

Part II

**Gluons, Scaling Violations, and
DGLAP**

Following Halzen & Martin §10.1–10.6

Chapter 5

The Dual Role of Gluons

5.1 Gluons as force carriers

From the perspective of symmetry, QCD is the gauge theory one obtains by promoting the global color- $SU(3)$ symmetry of free quarks to a local symmetry. The gauge field—the QCD analog of the photon in QED—is an eight-component vector field A_μ^a , with $a = 1, \dots, 8$ an index in the adjoint representation of $SU(3)$. The quanta of this field are the *gluons*. There are eight of them, one for each generator of the color group in the adjoint, and each gluon carries one unit of color charge and one unit of anticolor charge.

In contrast with QED, where the photon is electrically neutral and does not interact with itself, the gluons are *charged* under the very force they mediate. This is because $SU(3)$ is a non-abelian group: the generators T^a do not commute, and the structure constants f^{abc} that encode their commutators give rise to a three-gluon and a four-gluon vertex in the QCD Lagrangian. Schematically,

$$\mathcal{L}_{\text{QCD}} \supset -\frac{1}{4}(F_{\mu\nu}^a)^2, \quad F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g_s f^{abc} A_\mu^b A_\nu^c. \quad (5.1)$$

Squaring the field strength gives a kinetic term $\partial A \partial A$, plus cross-terms $\partial A \cdot f^{abc} AA$ that produce a three-gluon vertex, plus a quartic $(f^{abc} AA)^2$ that produces a four-gluon vertex.

5.1.1 Color factors

At every quark–gluon vertex, a factor $(T^a)_{ij}$ appears, where i and j are the fundamental indices of the incoming and outgoing quarks and a is the adjoint index of the gluon. The T^a are the eight 3×3 Gell-Mann matrices, normalised by $\text{Tr}(T^a T^b) = \frac{1}{2} \delta^{ab}$. Two identities will recur obsessively in what follows:

$$\sum_a (T^a T^a)_{ij} = C_F \delta_{ij}, \quad C_F = \frac{N_c^2 - 1}{2N_c} = \frac{4}{3}, \quad (5.2)$$

$$f^{acd} f^{bcd} = C_A \delta^{ab}, \quad C_A = N_c = 3, \quad (5.3)$$

$$\text{Tr}(T^a T^b) = T_R \delta^{ab}, \quad T_R = \frac{1}{2}. \quad (5.4)$$

C_F is the quadratic Casimir of the fundamental representation; C_A is the quadratic Casimir of the adjoint; T_R is the normalisation constant. These three numbers are the “color factors” of QCD, and we will see them explicitly in the splitting functions (and in any Compton-scattering problem set).

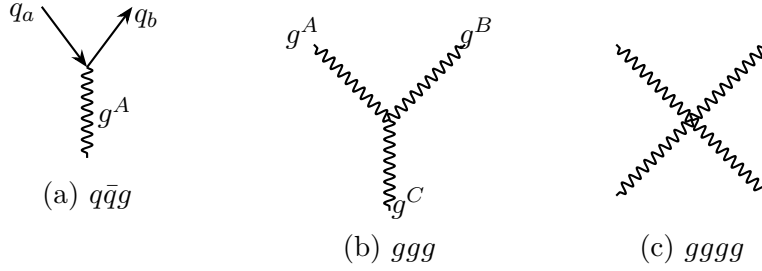


Figure 5.1: The three vertices of QCD: (a) the quark–gluon vertex, analogous to the QED vertex; (b) the three-gluon vertex, unique to non-abelian gauge theories; (c) the four-gluon vertex, also unique. The existence of (b) and (c) is the technical reason QCD is asymptotically free.

Compton color factor in QCD.

Consider the QCD process $q + g \rightarrow q + \gamma$ —the quark-gluon Compton scattering that appears in pN prompt-photon production. The amplitude has two diagrams: the gluon absorbed on the incoming quark (s-channel) and on the outgoing quark (u-channel). In each diagram, one quark–gluon vertex contributes a factor T^a . When we square and sum over the initial quark color (averaging by $1/N_c$) and over the final gluon color, we get

$$\frac{1}{N_c \cdot 8} \sum_{a,i,j} (T^a)_{ij} (T^a)_{ji} = \frac{1}{8N_c} \text{Tr}(T^a T^a) = \frac{1}{8N_c} \cdot 8 \cdot \frac{C_F}{1} \cdot \frac{1}{1} \cdot N_c \quad (5.5)$$

Wait, let me redo this. The correct evaluation is: $\sum_a (T^a T^a)_{ii} = C_F N_c$ (using $\sum_a T^a T^a = C_F \mathbf{1}$ and taking the trace), so that the squared-amplitude color factor is

$$\frac{1}{N_c} \sum_i \sum_a (T^a T^a)_{ii} = \frac{1}{N_c} \cdot C_F N_c = C_F = \frac{4}{3}. \quad (5.6)$$

(Note the $1/N_c$ is from averaging over the initial quark color; we do *not* average over the incoming gluon color because $g \rightarrow \text{photon}$ is trivial in the color index.) The same factor of $C_F = 4/3$ shows up in the DGLAP splitting function P_{qq} , for exactly the same reason: at the end of the day it is always the square of a single quark-gluon vertex, color-averaged.

5.1.2 Running coupling and asymptotic freedom

All of particle physics is dimensionally wrong unless we admit from the start that coupling constants “run” with energy scale. In QED, the fine-structure constant is $\alpha(0) = 1/137$ at zero momentum transfer but slowly grows as one probes smaller distances—it is about $1/128$ at the Z mass. This is charge screening: vacuum fluctuations into virtual electron-positron pairs partially shield the bare electric charge, so that as we probe closer to the source, the observed charge grows.

In QCD, quark loops do the same thing, but *gluon loops* do the opposite: because gluons carry color, they can “antiscreen” the color charge. For any number of quark flavors $n_f < 16$, the gluon contribution wins, and the coupling $\alpha_s(Q^2)$ *decreases* at short distances. To leading order,

$$\alpha_s(Q^2) = \frac{12\pi}{(33 - 2n_f) \ln(Q^2/\Lambda_{\text{QCD}}^2)} \quad (5.7)$$

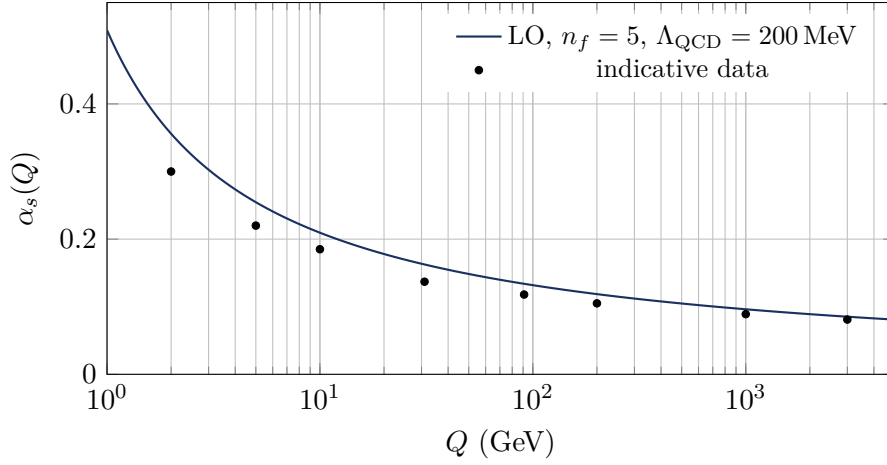


Figure 5.2: The running of $\alpha_s(Q)$ from Eq. (5.7). Asymptotic freedom is the statement that the curve slopes down, not up. At $Q \sim 1$ GeV, $\alpha_s \sim 0.3$ and perturbation theory starts to wobble; at $M_Z \sim 91$ GeV, $\alpha_s \approx 0.118$ and perturbation theory is firmly under control.

where $\Lambda_{\text{QCD}} \approx 200$ MeV is a dynamically generated scale—the one scale in QCD, since the theory has no dimensionful parameters in its Lagrangian (neglecting the tiny quark masses). The denominator $33 - 2n_f = 21$ for $n_f = 6$, so the coupling shrinks by $1/\ln$ at high Q^2 .

The physical consequence is that *perturbation theory is applicable to strong-interaction processes at high enough Q^2* . This is why we are allowed to write down Feynman diagrams for $\gamma^*q \rightarrow qg$ and take the results seriously: $\alpha_s(Q^2)$ is small at large Q^2 , and the series in α_s converges. At $Q^2 \sim \Lambda_{\text{QCD}}^2$, the coupling is of order unity, perturbation theory fails, and all the rich nonperturbative physics of confinement, chiral-symmetry breaking, and the hadron spectrum takes over.

5.2 Gluons as partons

Here is the second hat that gluons wear. We already know from the momentum sum rule (Eq. 4.10) that roughly half of the proton’s momentum is carried by electrically neutral partons. These neutral partons are gluons. So the parton distribution functions of the proton include, in addition to quark and antiquark distributions, a *gluon distribution* $g(x)$ —the probability density for finding a gluon with momentum fraction x .

The gluon PDF is real and measurable, and it contributes to many processes. In DIS itself, the gluon does not contribute at leading order (the photon doesn’t couple to gluons), but it enters at order α_s through the subprocess $\gamma^*g \rightarrow q\bar{q}$ and thereby influences the Q^2 evolution of F_2 . In pp collisions, many of the most important cross sections (Higgs production, $t\bar{t}$ production, direct photon production) are dominated by gluon-initiated subprocesses, and the gluon PDF enters directly.

Key idea

In QCD, gluons play two roles at once. (i) They are the carriers of the strong force between colored particles, exchanged along internal lines of Feynman diagrams and responsible for the binding of hadrons. (ii) They are partons in their own right, appearing on external lines of hard subprocesses, carrying momentum, and being described by a probability distribution $g(x, Q^2)$ that obeys the same DGLAP evolution as the quark distributions.

This dual role has no analog in QED (well—almost none; see the margin). In QED the photon is electrically neutral and does not itself contribute partonically at leading order to anything. A quark, by contrast, *does* carry color, and so do the gluons themselves, so the color analog of the photon structure function is nontrivial from the start.

Margin note

There *is* a QED analog, and it is pretty. At very high energies, a relativistic electron is partially dressed by its own collinear photon field, and this dressing can be organised into a “photon parton distribution inside the electron,” $f_{\gamma/e}(x)$. The probability of finding a photon carrying fraction x of the electron’s momentum is approximated by the Weizsäcker–Williams equivalent-photon formula, which you may recognise as a close relative of $P_{qq} \ln Q^2$. See H&M § 10.9 (optional).

Chapter 6

Embedding γ^* –Parton Processes in DIS

6.1 The “hat” notation

Almost every calculation in QCD phenomenology is a convolution of two things: a short-distance, perturbatively computable “hard subprocess” at the parton level, and one or more long-distance, nonperturbative parton distributions and/or fragmentation functions at the hadron level. The two levels live in two different worlds of kinematics, and it is convenient to put hats on the parton-level variables to distinguish them from the hadron-level variables.

For a $2 \rightarrow 2$ partonic subprocess $a(p_a) + b(p_b) \rightarrow c(p_c) + d(p_d)$ with $p_a = x_a P_A$ and $p_b = x_b P_B$, the *hadronic* Mandelstam variables are

$$s = (P_A + P_B)^2, \quad t = (P_A - p_c)^2, \quad u = (P_A - p_d)^2, \quad (6.1)$$

and the *partonic* ones (carrying hats) are

$$\hat{s} = (p_a + p_b)^2 = x_a x_b s, \quad \hat{t} = (p_a - p_c)^2, \quad \hat{u} = (p_a - p_d)^2, \quad (6.2)$$

with $\hat{s} + \hat{t} + \hat{u} = \sum m_i^2 \rightarrow 0$ for massless partons.

The key identity $\hat{s} = x_a x_b s$ is used *constantly* in parton-level calculations. In the context of DIS, the lepton is not a parton of a hadron, so only one momentum fraction appears: $\hat{s} = x s$ for the $\gamma^* q$ subprocess with P_A the proton. In the context of Drell–Yan production ($q\bar{q} \rightarrow \gamma^* \rightarrow \ell^+ \ell^-$), both incoming particles are partons of hadrons and we need two fractions: $\hat{s} = x_1 x_2 s$. This is the formula your professor was pointing at on the relevant page of H&M.

Key idea

Whenever a hatted variable appears, you are looking at the parton-level subprocess. The relation to the hadronic level is almost always $\hat{s} = x s$ for DIS-like processes or $\hat{s} = x_1 x_2 s$ for hadron-hadron collisions. Internalise this and you will not get confused.

6.2 Embedding: the general convolution

A generic inclusive cross section for a hard process on a proton can be written as a convolution:

$$d\sigma^{\text{hadr}}(P, k) = \sum_i \int_0^1 d\xi f_{i/p}(\xi, \mu^2) d\hat{\sigma}_i(\xi P, k; \mu^2), \quad (6.3)$$

where the sum runs over parton species and μ^2 is a factorisation scale, typically chosen equal to the hard scale of the process. The parton distribution $f_{i/p}(\xi, \mu^2)$ is the probability of finding parton i inside the proton with momentum fraction ξ when one resolves distances of order $1/\mu$. The partonic cross section $d\hat{\sigma}_i$ is computed perturbatively from Feynman diagrams.

Equation (6.3) is the *factorisation theorem*. It is not obvious; it was conjectured by Drell, Yan, and others in the 1970s and proven (for a wide class of processes) by Collins, Soper, and Sterman in the 1980s. The statement is that the nonperturbative physics of the proton’s bound-state structure *decouples* from the perturbative physics of the hard scattering, and the two can be separated into universal, process-independent PDFs and process-dependent, perturbative hard cross sections.

Margin note

“Factorisation is a theorem.” Whenever you see a convolution of a PDF with a partonic cross section, remember that this factorised form is not just a plausible ansatz—it is provable from QCD, with a specific set of conditions (inclusive enough in the final state, $Q^2 \gg \Lambda_{\text{QCD}}^2$, no soft gluon exchanges between initial and final hadrons, etc.). Violations of factorisation are a subtle and active subject (small- x saturation, ultraperipheral collisions, ...).

6.3 The leading-order DIS subprocess

For DIS, the leading-order partonic subprocess is just the elastic scattering of a virtual photon off a point quark: $\gamma^*(q) + q(p) \rightarrow q(p')$. There is only one diagram, and it is essentially the Mott diagram with the quark charge e_q . The partonic cross section, after the usual leptonic-tensor contraction, is

$$\frac{d\hat{\sigma}_q^{LO}}{d\hat{x} dQ^2} = \frac{2\pi\alpha^2 e_q^2}{Q^4} \left[1 + (1 - \hat{y})^2 \right] \delta(\hat{x} - 1), \quad (6.4)$$

where $\hat{x} = Q^2/(2p \cdot q)$ is the partonic Bjorken variable (note the distinction from the hadronic $x = Q^2/(2P \cdot q)$; they are related by $\hat{x} = x/\xi$). The delta function enforces on-shellness of the struck quark, collapsing the convolution of Eq. (6.3) to yield

$$F_2^{LO}(x) = \sum_q e_q^2 x f_q(x), \quad (6.5)$$

reproducing the parton-model result. At this order, scaling is exact.

The interesting physics begins at NLO—the first $\mathcal{O}(\alpha_s)$ correction—where the struck quark can radiate a gluon. This is where scaling gets broken and the Altarelli–Parisi equation is born.

Chapter 7

The Parton Model Revisited

7.1 Leading order and higher orders

We now have a crisp way of thinking about the parton model. Define a set of partonic cross sections $\hat{\sigma}^{(n)}$, where n counts the number of extra powers of α_s beyond the leading electromagnetic interaction. Then the full DIS structure function is

$$F_2(x, Q^2) = \sum_i \int d\xi f_i(\xi, \mu^2) \left[\hat{F}_2^{(0)}(x/\xi, Q^2/\mu^2) + \alpha_s \hat{F}_2^{(1)}(x/\xi, Q^2/\mu^2) + \mathcal{O}(\alpha_s^2) \right]. \quad (7.1)$$

The first term is the naïve parton-model contribution—exact scaling, trivial x -convolution with a delta function. The second term is the one-loop QCD correction; it introduces a nontrivial x -dependence of the partonic coefficient function (not just a delta) and, crucially, a *logarithm* of Q^2/μ^2 . It is this logarithm that breaks scaling and whose resummation (via the AP equation) gives the predicted pattern of scaling violations.

7.2 What a single order- α_s gluon emission looks like

The $\mathcal{O}(\alpha_s)$ correction to $\gamma^* q \rightarrow qX$ has two categories of diagrams. The first is “real” emission: the struck quark radiates a gluon before or after the photon vertex, so the final state contains a quark and a gluon. The second is “virtual” emission: the quark emits and reabsorbs a gluon in a loop, so the final state still contains only a quark. Both contribute at the same order in α_s , and both must be combined to get a finite result.

The key computational observation is that each of the real emission diagrams has a propagator $1/(p_{\text{virtual}})^2$, which in the limit of collinear emission becomes very small (i.e., the propagator becomes very large). Parametrising the gluon momentum as $k^\mu = (1-z)p^\mu + k_\perp^\mu + (\text{small})$, where z is the momentum fraction of the final quark, one finds that the virtuality of the intermediate quark scales as $p_{\text{virt}}^2 \sim k_\perp^2/(1-z)$, so the propagator squared contributes $1/k_\perp^4$ to the amplitude squared. When we integrate over the phase space of the emitted gluon, we find

$$\int d^2 k_\perp \frac{1}{k_\perp^4} \cdot k_\perp^2 \sim \int \frac{dk_\perp^2}{k_\perp^2} = \ln \frac{k_{\perp, \text{max}}^2}{k_{\perp, \text{min}}^2}. \quad (7.2)$$

The upper limit of the $k_\perp^2$ integral is Q^2 (the hard scale), and the lower limit is set by some low scale μ_0^2 (which we will regulate shortly). The result is a *collinear logarithm* $\ln(Q^2/\mu_0^2)$ in the partonic cross section, multiplied by a function of z called the *splitting function* $P_{qq}(z)$.

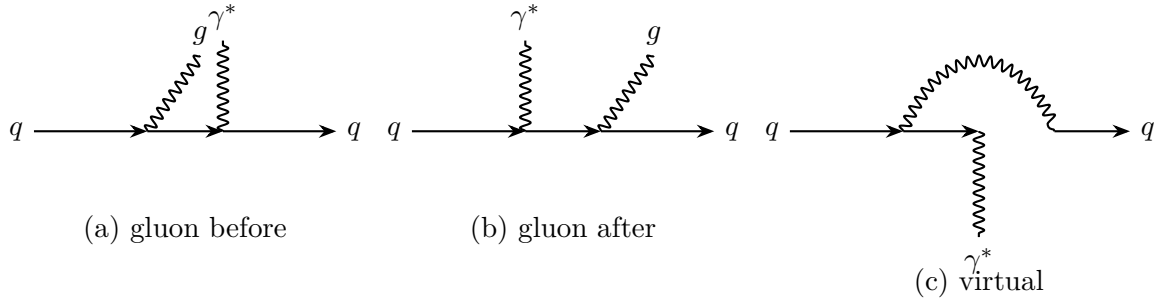


Figure 7.1: Order- α_s corrections to $\gamma^* q \rightarrow q(g)$. The real emission diagrams (a) and (b) add a final-state gluon; the virtual diagram (c) has the same final state as the Born process. All must be summed to obtain a finite, physical cross section.

This is the origin of scaling violations: the collinear logarithm is not an artefact—it is real, it is Q^2 -dependent, and it is the leading effect at NLO.

Chapter 8

The Gluon Emission Cross Section

8.1 Setup

Let us be more precise. Consider the subprocess $\gamma^*(q) + q(p) \rightarrow q(p') + g(k)$. Let z be the fraction of the initial quark's momentum that survives in the final-state quark: $p' \approx zp$. Then the gluon carries fraction $(1 - z)$ plus a transverse component $k_\perp$, so its four-momentum is approximately

$$k^\mu \approx (1 - z)p^\mu + k_\perp^\mu + (\text{terms of order } k_\perp^2/[(1 - z)p]) \quad (8.1)$$

in the light-cone frame where p points along the positive z -axis. The virtuality of the intermediate quark—call it p^* —is

$$p^{*2} = -\frac{k_\perp^2}{1 - z}. \quad (8.2)$$

This is the crucial kinematic quantity. It vanishes in the collinear limit $k_\perp \rightarrow 0$, making the propagator $1/(p^{*2} - m^2) \rightarrow \infty$, and generating the logarithmic divergence we mentioned.

8.2 The splitting function P_{qq}

A careful calculation of the amplitude squared, averaging over initial spins and colors, summing over final spins and colors, and integrating over the gluon phase space in the collinear limit, gives

$$\int_{\mu_0^2}^{Q^2} \frac{dk_\perp^2}{k_\perp^2} \frac{\alpha_s}{2\pi} P_{qq}(z), \quad (8.3)$$

with

$$P_{qq}(z) = C_F \frac{1 + z^2}{1 - z}. \quad (8.4)$$

The function $P_{qq}(z)$ is the leading-order *quark-to-quark splitting function*: it is the probability per unit $\ln k_\perp^2$ that a quark splits into a quark carrying fraction z of the parent momentum (and a gluon carrying fraction $1 - z$).

The color factor $C_F = 4/3$ in Eq. (8.4) comes directly from squaring the quark–gluon vertex: each vertex gives a factor T^a , and summing the outer product $T^a T^a$ over a gives $C_F \mathbf{1}$. Averaging over the initial quark color ($1/N_c$) and summing over the final gluon color ($N_c^2 - 1$) gives the combination $(N_c^2 - 1)/(2N_c) = C_F$.

8.2.1 The soft singularity and the plus prescription

As $z \rightarrow 1$, the splitting function (8.4) diverges like $1/(1-z)$: this is the *soft gluon singularity*, which physically says that emitting an arbitrarily soft gluon is an arbitrarily likely event. The divergence is not physical—it is cancelled by the virtual correction, which has an identical but opposite singularity at $z = 1$. When one combines the real and virtual contributions, the infinity cancels but leaves behind a finite, regular function of z , plus a delta function at $z = 1$ coming from the “endpoint” part of the virtual correction. The regularisation is conventionally done using the *plus prescription*: for any test function $g(z)$,

$$\int_0^1 dz [f(z)]_+ g(z) \equiv \int_0^1 dz f(z) [g(z) - g(1)], \quad (8.5)$$

which means that the integral is regulated by subtracting the endpoint. In the plus prescription, the fully-regulated splitting function is

$$P_{qq}(z) = C_F \left[\frac{1+z^2}{(1-z)_+} + \frac{3}{2} \delta(1-z) \right]. \quad (8.6)$$

The $\frac{3}{2}\delta(1-z)$ term comes from the virtual correction, and it is what makes the number of quarks ($\int q dx$) conserved under Q^2 -evolution, as baryon-number conservation requires.

8.3 The other splitting functions

The same calculation can be done for the other parton-level splittings that appear at $\mathcal{O}(\alpha_s)$. There are four of them: $q \rightarrow q$, $q \rightarrow g$, $g \rightarrow q$, and $g \rightarrow g$. (By “ $a \rightarrow b$ ” here we mean “parton a splits into parton b plus something else”—i.e., b is the “seen” daughter and carries fraction z of the parent.) The leading-order results are:

$$P_{qq}(z) = C_F \left[\frac{1+z^2}{(1-z)_+} + \frac{3}{2} \delta(1-z) \right], \quad (8.7)$$

$$P_{gq}(z) = C_F \frac{1+(1-z)^2}{z}, \quad (8.8)$$

$$P_{qg}(z) = T_R [z^2 + (1-z)^2], \quad (8.9)$$

$$P_{gg}(z) = 2C_A \left[\frac{z}{(1-z)_+} + \frac{1-z}{z} + z(1-z) \right] + \frac{11C_A - 4n_f T_R}{6} \delta(1-z). \quad (8.10)$$

The color factors reveal the origin of each process: P_{qq} and P_{gq} involve a single quark-gluon vertex (color factor C_F); P_{qg} involves a gluon splitting into a quark-antiquark pair at a quark-gluon-antiquark vertex (normalisation T_R times the number of final-state quark flavors, which we have factored into the definition); P_{gg} involves the three-gluon vertex (color factor C_A).

8.3.1 Probabilistic interpretation

Each splitting function $P_{ij}(z)$ has a natural interpretation as a probability density per unit $\ln Q^2$ for the splitting $i \rightarrow j + \text{anything}$ with daughter momentum fraction z . This is the most physical way to think of the AP equation: at each step in “time” $t = \ln Q^2$, a parton undergoes infinitesimal branching. The branching probabilities are the splitting functions. The parton content of a hadron is then a Markov process in t , governed by the splitting functions as a transition kernel.

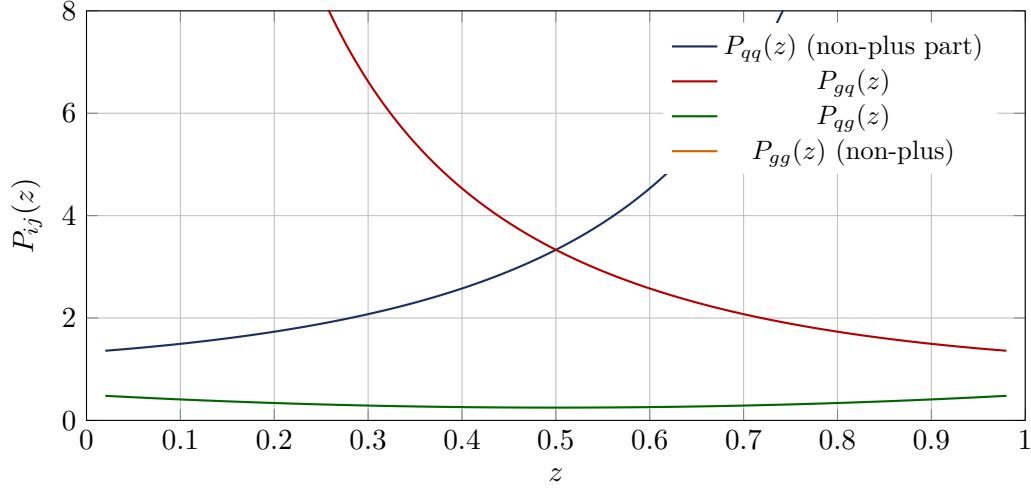


Figure 8.1: The four QCD splitting functions at leading order. Notice the soft singularities at $z = 1$ in P_{qq} and P_{gg} , and at $z = 0$ in P_{gq} and P_{gg} (the “soft gluon” from the radiated side). P_{qg} has no soft singularity at all, and is symmetric around $z = 1/2$.

Explicit color derivation of C_F .

The quark-gluon vertex gives a factor $(T^a)_{ij}$ for a quark of color i emitting a gluon of color a and becoming a quark of color j . Squaring and summing over final gluon color and final quark color, while *averaging* over the initial quark color (multiplying by $1/N_c$), we find

$$\frac{1}{N_c} \sum_{i,j,a} (T^a)_{ij} (T^a)_{ji}^* = \frac{1}{N_c} \sum_i \sum_a (T^a T^a)_{ii}. \quad (8.11)$$

Using the Casimir identity $\sum_a T^a T^a = C_F \mathbf{1}$ (with $C_F = (N_c^2 - 1)/(2N_c)$), the sum over i gives N_c , and we obtain

$$\frac{1}{N_c} \cdot N_c C_F = C_F = \frac{4}{3}. \quad (8.12)$$

The same factor $C_F = 4/3$ appears in the Altarelli–Parisi splitting function P_{qq} , and in the QCD correction to $R_{e^+e^-}$, and in countless other places. *It is always the same number, and it always comes from the same color identity.*

8.4 The coefficient function and factorisation

Assembling the real and virtual corrections, one finds that the structure function F_2 at NLO takes the schematic form

$$F_2(x, Q^2) = \sum_q e_q^2 x \left[f_q(x) + \frac{\alpha_s}{2\pi} \int_x^1 \frac{dz}{z} f_q(x/z) \left\{ P_{qq}(z) \ln(Q^2/\mu_0^2) + C_q(z) \right\} \right] + (\text{gluon piece}). \quad (8.13)$$

Here μ_0^2 is an infrared regulator that we have been sloppy about, $C_q(z)$ is a finite coefficient function, and the logarithm is the ultraviolet part of the $k_\perp^2$ integral. The logarithm depends on the regulator,

which is unphysical. The trick is to absorb it into a redefinition of the PDF:

$$f_q(x, Q^2) \equiv f_q(x) + \frac{\alpha_s}{2\pi} \ln(Q^2/\mu_0^2) \int_x^1 \frac{dz}{z} f_q(x/z) P_{qq}(z) + \mathcal{O}(\alpha_s^2). \quad (8.14)$$

This *factorisation* absorbs the infrared (μ_0) divergence into a scale-dependent PDF and leaves behind a finite, Q^2 -dependent coefficient function. Differentiating Eq. (8.14) with respect to $\ln Q^2$ gives the Altarelli–Parisi equation.

Chapter 9

Scaling Violations and the Altarelli–Parisi Equation

9.1 Derivation of DGLAP from factorisation

Take Eq. (8.14), apply $\partial/\partial \ln Q^2$ to both sides, and work to leading order in α_s . The left-hand side is $\partial f_q/\partial \ln Q^2$, and the right-hand side gives the convolution

Altarelli–Parisi (DGLAP) equation

$$\frac{\partial f_q(x, Q^2)}{\partial \ln Q^2} = \frac{\alpha_s(Q^2)}{2\pi} \int_x^1 \frac{d\xi}{\xi} P_{qq}\left(\frac{x}{\xi}\right) f_q(\xi, Q^2). \quad (9.1)$$

This is the *Altarelli–Parisi equation*, first derived in its modern form by Altarelli and Parisi in 1977, and independently (in different language) by Dokshitzer in 1977 and by Gribov and Lipatov in the early 1970s. It is universally known nowadays as the *DGLAP equation*.

Equation (9.1) is a first-order integro-differential equation for the parton density as a function of Q^2 . To make it into a *predictive* equation, you need an initial condition—the PDF at some reference scale Q_0^2 —and you need to solve for $f_q(x, Q^2)$ at all other scales. The initial condition must come from experiment; QCD alone cannot compute PDFs at any given scale. But once you have them at one scale, QCD tells you what they are at every other scale.

9.2 Physical interpretation: branching with “time” $t = \ln Q^2$

Define $t \equiv \ln(Q^2/Q_0^2)$. Then the AP equation becomes

$$\frac{\partial f_q(x, t)}{\partial t} = \frac{\alpha_s}{2\pi} \int_x^1 \frac{d\xi}{\xi} P_{qq}(x/\xi) f_q(\xi, t). \quad (9.2)$$

This has the structure of a linear evolution equation in “time” t , driven by an integral kernel. Physically, it is a forward Kolmogorov equation for a branching process: as we increase t (i.e., as we probe at higher Q^2), the parton content evolves by a series of infinitesimal branchings, each governed by the splitting probability P_{qq} . A parent quark at momentum fraction ξ branches into a daughter quark at a lower fraction $x < \xi$ (and a gluon at fraction $\xi - x$), with probability density $P_{qq}(x/\xi)$.

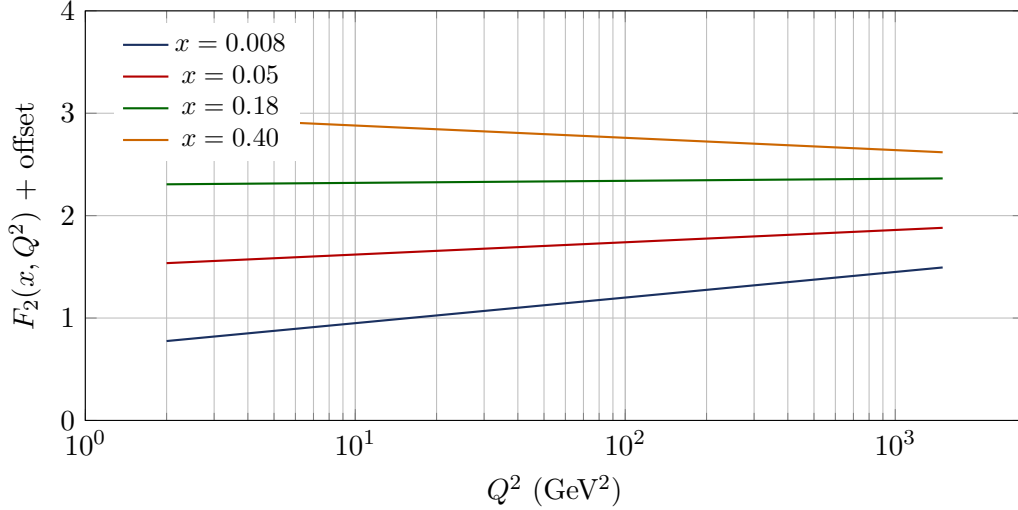


Figure 9.1: Schematic pattern of scaling violations: $F_2(x, Q^2)$ as a function of Q^2 at fixed x . At small x (blue), F_2 grows with Q^2 ; at large x (orange), F_2 decreases. At an intermediate value $x \approx 0.15\text{--}0.2$, F_2 is nearly independent of Q^2 —this is where Bjorken scaling is “least violated.” Vertical offsets are for display.

The picture explains the direction of scaling violations qualitatively. At large x (the valence peak), the branching sends momentum from high x to low x , depleting the large- x region and enriching the small- x region. Therefore $f(x, Q^2)$ *decreases* with Q^2 at large x , and *increases* with Q^2 at small x . The crossover point is around $x \approx 0.15$. This qualitative prediction is exactly borne out by the HERA data.

9.3 The coupled system with gluons

Equation (9.1) is only half the story. The quark distribution can evolve because of quark-to-quark splitting, which is proportional to the quark density itself, *and* because of gluon-to-quark splitting (the $g \rightarrow q\bar{q}$ process), which is proportional to the gluon density. Similarly, the gluon distribution evolves via both $g \rightarrow g$ and $q \rightarrow g$ splittings. The coupled system is:

Coupled DGLAP

$$\frac{\partial q_i(x, t)}{\partial t} = \frac{\alpha_s(t)}{2\pi} \int_x^1 \frac{d\xi}{\xi} \left[P_{qq}(x/\xi) q_i(\xi, t) + P_{qg}(x/\xi) g(\xi, t) \right], \quad (9.3)$$

$$\frac{\partial g(x, t)}{\partial t} = \frac{\alpha_s(t)}{2\pi} \int_x^1 \frac{d\xi}{\xi} \left[\sum_j P_{gq}(x/\xi) q_j(\xi, t) + P_{gg}(x/\xi) g(\xi, t) \right]. \quad (9.4)$$

These are the full leading-order DGLAP equations. The index i runs over all quarks *and* antiquarks (six species each, though only a subset is usually active). The sum in the gluon equation runs over all quark flavors.

9.3.1 Singlet and nonsinglet decomposition

The coupled system is easier to solve if one decomposes the quark distributions into a singlet and nonsinglet basis:

$$\text{singlet:} \quad \Sigma(x, t) = \sum_i [q_i(x, t) + \bar{q}_i(x, t)], \quad (9.5)$$

$$\text{nonsinglet:} \quad q_{\text{NS}}^{(ij)}(x, t) = q_i(x, t) - q_j(x, t) \text{ or } q_i(x, t) - \bar{q}_i(x, t). \quad (9.6)$$

The nonsinglet combinations evolve through P_{qq} alone—the gluon drops out, because it contributes equally to all quark flavors and therefore cancels in any flavor difference. The singlet couples to the gluon through P_{qg} and P_{gq} . This decomposition reduces the coupled system (9.3)–(9.4) to two decoupled equations (for nonsinglet and for the Σ – g pair), which is a major practical simplification.

9.3.2 Mellin transforms: turning convolutions into products

The convolution structure of the DGLAP equations is best handled by a Mellin transform. Define, for any function $f(x)$,

$$\tilde{f}(N) \equiv \int_0^1 dx x^{N-1} f(x). \quad (9.7)$$

A convolution of the form $\int (d\xi/\xi) P(x/\xi) f(\xi)$ becomes, in Mellin space, simply the product $\tilde{P}(N)\tilde{f}(N)$. Thus the AP equation in Mellin space is

$$\frac{\partial \tilde{f}_q(N, t)}{\partial t} = \frac{\alpha_s(t)}{2\pi} \tilde{P}_{qq}(N) \tilde{f}_q(N, t), \quad (9.8)$$

which is an ordinary linear differential equation in t . Its solution is an exponential:

$$\tilde{f}_q(N, t) = \tilde{f}_q(N, 0) \exp \left[\frac{\tilde{P}_{qq}(N)}{2\pi} \int_0^t \alpha_s(t') dt' \right]. \quad (9.9)$$

For small N , this grows rapidly in t (the small- x rise at large Q^2); for larger N , it falls (the large- x depletion). The full x -space solution is recovered by an inverse Mellin transform.

9.3.3 The Mellin moments of P_{qq}

For the curious reader, the Mellin moments of the plus-prescription splitting function P_{qq} are

$$\tilde{P}_{qq}(N) = C_F \left[\frac{3}{2} - \frac{1}{N(N+1)} - 2 \sum_{k=2}^N \frac{1}{k} \right], \quad (9.10)$$

which for $N = 1$ vanishes. This is not an accident: it is the quark number sum rule. The integral $\int_0^1 q(x, t) dx$ is the total number of quarks, which is conserved by evolution. In Mellin space, this is the statement that $\tilde{q}(N = 1, t)$ is t -independent, which follows from $\tilde{P}_{qq}(1) = 0$.

9.4 Solving DGLAP numerically

In practice, the way you solve DGLAP is:

1. **Input.** Specify the PDFs at a reference scale Q_0^2 , usually 1–2 GeV², from a fit to data.

2. **Evolve up.** Use a numerical method (Runge–Kutta, or something more sophisticated based on Mellin or Laguerre-polynomial bases) to integrate the coupled DGLAP equations forward in $t = \ln Q^2$.
3. **Compare with data.** At each Q^2 of experimental interest, compute the observable (say F_2^p) and compare with measurements.
4. **Iterate.** Adjust the input PDFs at Q_0^2 until the χ^2 with respect to all the data is minimised. This is a global fit—the modern PDFs (CT, MSHT, NNPDF) are all produced in this way.

The resulting agreement between theory and experiment is, quite literally, one of the greatest quantitative successes of any physical theory. Across five orders of magnitude in Q^2 and several orders of magnitude in x , the HERA data on $F_2(x, Q^2)$ are described by DGLAP evolution from a common initial condition at the percent level.

Margin note

This extraordinary agreement is sometimes called “the precision test of QCD at the parton level.” Note what is being tested: not the parton distributions themselves (those are input), but the evolution kernel—the splitting functions—which *is* predicted perturbatively by QCD from first principles.

Chapter 10

Including Gluon Pair Production

10.1 Why gluon pair production matters

We have seen that scaling violations come from gluon emission ($q \rightarrow qg$) inducing a collinear log. There is a parallel effect from gluon *absorption*: a gluon in the proton splits into a quark and an antiquark, and the photon scatters off one of them. Schematically, $\gamma^* + g \rightarrow q + \bar{q}$. This process contributes to F_2 at order α_s exactly as the real-emission correction does, and it must be included for consistency.

The physical picture is that the gluon inside the proton is, at finite $k_\perp$, indistinguishable from a $q\bar{q}$ pair. From the photon's point of view, the gluon “appears” as a quark pair (whichever one has the right charge to couple to the photon). This is how the photon effectively sees the gluon distribution, even though it does not couple to gluons directly.

10.2 Modifying the DGLAP equation

Including the $\gamma^*g \rightarrow q\bar{q}$ process, the evolution of the quark distribution picks up an extra term proportional to the gluon distribution:

$$\frac{\partial q_i(x, t)}{\partial t} = \frac{\alpha_s}{2\pi} \int_x^1 \frac{d\xi}{\xi} \left[P_{qq}(x/\xi) q_i(\xi, t) + P_{qg}(x/\xi) g(\xi, t) \right], \quad (10.1)$$

with P_{qg} as in Eq. (8.9). The factor of $T_R = \frac{1}{2}$ in P_{qg} comes from $\text{Tr}(T^a T^b) = \frac{1}{2} \delta^{ab}$: when the gluon splits into a quark-antiquark pair, the color-index sum gives exactly this factor.

Similarly, the gluon distribution evolves through its own $g \rightarrow g$ splittings (the P_{gg} contribution) and through $q \rightarrow g$ radiation (the P_{gq} contribution). All together, we recover the coupled system (9.3)–(9.4).

10.3 Momentum conservation and the sum rules of DGLAP

Two elegant sum rules follow from the structure of the splitting functions.

Quark number. The total number of quarks of each flavor is conserved by evolution:

$$\int_0^1 P_{qq}(z) dz = 0, \quad (10.2)$$

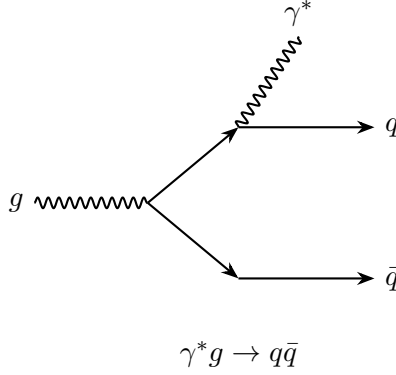


Figure 10.1: Gluon-initiated DIS subprocess at $\mathcal{O}(\alpha_s)$. The gluon from the proton splits into a $q\bar{q}$ pair; the virtual photon scatters off one of them. This diagram is what feeds the sea-quark distribution through the P_{qg} splitting function.

where the integral is understood in the plus-prescription sense (since P_{qq} is a distribution). This is why the integral $\int (q - \bar{q}) dx$ —the baryon number—is Q^2 -independent.

Momentum conservation. The total momentum carried by all partons is conserved by evolution:

$$\int_0^1 dz z [P_{qq}(z) + P_{gq}(z)] = 0, \quad \int_0^1 dz z [2n_f P_{qg}(z) + P_{gg}(z)] = 0. \quad (10.3)$$

These are the momentum sum rules for quark- and gluon-initiated evolution, respectively. They correspond to the statement that the integral $\int x [\sum_i (q_i + \bar{q}_i) + g] dx = 1$ (the total momentum fraction carried by partons) is Q^2 -independent.

Key idea

The DGLAP equations conserve the quark number and the total parton momentum at every Q^2 . These are symmetries of QCD that must be preserved by any consistent evolution formula. You can use them as a sanity check whenever you write down or numerically solve DGLAP.

10.4 Beyond leading order

We have presented everything at *leading order* in α_s . The next-to-leading order (NLO) and next-to-next-to-leading order (NNLO) splitting functions are now known—the NLO ones were computed by Floratos, Ross, and Sachrajda; Curci, Furmanski, and Petronzio; and others in the late 1970s and early 1980s, and the NNLO ones by Moch, Vermaseren, and Vogt in 2004. Very recently (2023–2024), a substantial part of the N³LO splitting functions has been computed as well, and modern “N³LO PDFs” are available.

For LHC phenomenology at the percent level, NNLO is the standard, and many processes are computed at NNLO+NNLL (the extra *L*s referring to resummation of large logarithms from soft/collinear emissions). At the level of this course, LO is sufficient to understand the *ideas*.

Self-check

(1) Starting from the definition $f_q(x, Q^2)$ in Eq. (8.14), derive the AP equation (9.1) by applying $\partial/\partial \ln Q^2$. Where does the $\alpha_s(Q^2)$ dependence of the coefficient come from? (2) Verify that $\int_0^1 P_{qq}(z) dz = 0$ when P_{qq} is the full plus-prescription expression (8.7). (3) The first Mellin moment of $xP_{qq}(z)$ gives the evolution of the momentum-fraction integral $\int xq(x)dx$. Compute it, and check that it is cancelled by the xP_{gq} moment. (4) Make a qualitative plot of $\partial f_q/\partial t$ as a function of x , assuming a valence- peaked initial condition, and verify that the sign of the derivative is consistent with the depletion/enrichment picture in the text.

Part III

Fragmentation: What Comes Out of a Hadron

Chapter 11

Fragmentation Functions and Their Scaling Properties

11.1 Motivation: why we need another set of functions

Let me motivate this chapter with a practical puzzle. Imagine you are sitting in the control room at LEP in the late 1990s, watching e^+e^- collisions at $\sqrt{s} = 91$ GeV. Every few seconds, two beams crash together, the collision point lights up, and a detector records the energies and angles of everything that comes out. What you typically see is two back-to-back sprays of about 20 charged particles each — two “jets” of hadrons. You never see a bare quark; you see a shower of pions, kaons, and sometimes protons.

From the arguments of the previous parts of this book, we know what *ought* to be happening at the fundamental level. The e^+e^- pair annihilates into a virtual photon (or Z boson), which then materialises as a quark–antiquark pair. Those are the two “things” flying apart at the speed of light, one in each direction. But what we observe are hadrons, not quarks. Between the quark and the detector, something has to happen — the quark has to turn into hadrons.

The trouble is that this “something” is hopelessly non-perturbative. It happens at the confinement scale, where $\alpha_s \sim 1$, where Feynman diagrams are useless, and where the underlying dynamics involve the breaking of QCD flux tubes, the formation of $q\bar{q}$ pairs out of the vacuum, and all the messy business of hadronisation. We cannot calculate it from first principles. What we *can* do is admit defeat on the theoretical side, package all of our ignorance into a single universal function, measure it in one experiment, and then use it everywhere else. That universal function is the **fragmentation function** $D_i^h(z, Q^2)$.

Key idea

In the same way that we packaged our ignorance about *what is inside the proton* into parton distribution functions $f_i(x, Q^2)$, we package our ignorance about *how quarks turn into hadrons* into fragmentation functions $D_i^h(z, Q^2)$. The former is the “input” side of a hard scattering; the latter is the “output” side. Both are universal — measure them once, use them everywhere.

11.2 Definition, very carefully

Here is the definition, stated as precisely as we can without yet getting into the formal operator definitions. Imagine a parton of type i (say a u -quark) that has been produced by some hard process — it could be e^+e^- annihilation, it could be the struck quark in DIS, it could be a quark emerging from a $t\bar{t}$ event at the LHC. Watch that parton hadronise, and look for hadrons of type h (say π^+) in the resulting shower. For each such pion, record the fraction z of the parent parton's energy that it carries off. Now do this for many, many events, and histogram z .

The result is (up to a normalisation) the fragmentation function $D_i^h(z)$. More precisely, $D_i^h(z) dz$ is the *mean number* of hadrons of type h with energy fraction in $[z, z + dz]$ produced by a parent parton of type i . The key word here is “mean” — it is an average over many events, not a probability. A single quark can produce multiple pions of varying z , so the integral

$$\int_0^1 D_i^h(z) dz = \langle n_h \rangle_i$$

is the average multiplicity of h 's in a jet initiated by parton i , and that number is generally larger than 1.

Margin note

It is easy to confuse yourself by thinking of $D_i^h(z)$ as a probability. It is not, exactly — it is a density. What is normalised is not the number of pions but their total energy (see the energy sum rule below).

The variable z is, by definition, the fraction of the parent's energy (or longitudinal momentum, which is nearly the same in the high-energy limit) carried by the daughter hadron. In e^+e^- at c.m. energy $\sqrt{s}$, each quark has energy $\sqrt{s}/2$, so a pion of energy E_π has $z = 2E_\pi/\sqrt{s}$.

11.3 The cleanest experiment: $e^+e^- \rightarrow hX$

Let's start with the cleanest setting. In e^+e^- annihilation at $\sqrt{s} = Q$ well above any quark mass, the total cross section to hadrons is, to leading order in α_s ,

$$\sigma_{\text{tot}}(e^+e^- \rightarrow \text{hadrons}) = \frac{4\pi\alpha^2}{3Q^2} \sum_q e_q^2 \cdot N_c, \quad (11.1)$$

where $N_c = 3$ is the number of colours, e_q is the quark charge in units of $|e|$, and the sum is over quark flavours produced at that energy (so at LEP, u, d, s, c, b). This is just $R \cdot \sigma(e^+e^- \rightarrow \mu^+\mu^-)$ where

$$R \equiv \frac{\sigma(\text{hadrons})}{\sigma(\mu^+\mu^-)} = N_c \sum_q e_q^2.$$

The reason we bring this up is that the inclusive rate for producing a specific hadron h is, by the definition of D_q^h ,

$$\frac{1}{\sigma_{\text{tot}}} \frac{d\sigma(e^+e^- \rightarrow hX)}{dz} = \frac{\sum_q e_q^2 [D_q^h(z, Q^2) + D_{\bar{q}}^h(z, Q^2)]}{\sum_q e_q^2}. \quad (11.2)$$

On the left-hand side is a directly measurable quantity. On the right-hand side is the unknown D 's. By making this measurement at many values of z and many values of Q , an experimentalist *extracts* the fragmentation functions, just as DIS experimentalists extract PDFs.

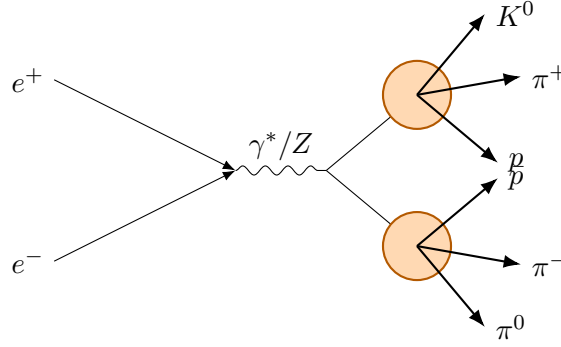


Figure 11.1: $e^+e^- \rightarrow q\bar{q} \rightarrow \text{hadrons}$. The hard process (annihilation through the virtual photon) is calculable in perturbative QED. Each of the outgoing quarks then hadronises — a non-perturbative step described by $D_q^h(z, Q^2)$ — producing a collimated spray of hadrons, a jet.

11.4 Sum rules: what the fragmentation functions *must* satisfy

Even without calculating any D from first principles, we can derive constraints that they must obey from basic conservation laws. These are the *sum rules*, and they are the fragmentation analogues of the sum rules we met for PDFs in Chapter 4.

The energy (momentum) sum rule. Imagine a parton i carrying energy E . It hadronises and produces a bunch of hadrons with energies $E_1, E_2, \dots$. The sum of the daughter energies must equal the parent’s energy (strictly, the longitudinal-momentum version holds; at high boost they’re identical). Averaging over many events and summing over all possible hadron species h ,

$$\sum_h \int_0^1 z D_i^h(z, Q^2) dz = 1.$$

In words: if you sum the fractional energies carried off by every hadron in the jet, you get 1, because energy is conserved. The left-hand side is a sum of *mean* energy fractions — this is a statement about the average jet.

Where did the z come from? Here is a small subtlety worth pausing on. We wrote $\int z D_i^h dz$ and not just $\int D_i^h dz$. Why? Because $D_i^h(z) dz$ is the mean *number* of hadrons in the interval, and each such hadron carries energy fraction z . So the mean total energy fraction carried by h ’s is $\int z D_i^h(z) dz$. Summing over h and demanding the total is 1 gives the energy sum rule.

The multiplicity sum rule. Now suppose we don’t weight by energy, but just count hadrons. Then

$$\langle n_h \rangle_i(Q^2) = \int_{z_{\min}}^1 D_i^h(z, Q^2) dz$$

is the average number of h ’s produced by parton i . This is not constrained to 1; it’s a measurable quantity whose value depends on i , h , and Q^2 . The cutoff $z_{\min}$ is there because D diverges like $1/z$ as $z \rightarrow 0$ (physically, jets produce many soft hadrons), and so the integral would formally diverge without some lower limit set by kinematics (typically the minimum-observable hadron energy).

Charge-conjugation and isospin. In the limit where we ignore small differences between u - and d -quarks and between quarks and antiquarks, we have relations like

$$\begin{aligned} D_u^{\pi^+}(z) &= D_d^{\pi^+}(z) = D_d^{\pi^-}(z) = D_{\bar{u}}^{\pi^-}(z) \quad (\text{favoured}), \\ D_u^{\pi^-}(z) &= D_{\bar{d}}^{\pi^-}(z) = D_d^{\pi^+}(z) = D_{\bar{u}}^{\pi^+}(z) \quad (\text{unfavoured}). \end{aligned}$$

The physical content is this: a $\pi^+ = u\bar{d}$ contains a valence u -quark, so a u -quark fragmenting into a π^+ is “favoured” — the valence quark just needs to pick up a $\bar{d}$ from the vacuum. By contrast, a u -quark fragmenting into a $\pi^- = \bar{u}d$ has to produce *both* a $\bar{u}$ and a d from the vacuum — the original u doesn’t appear in the final pion — so this process is “unfavoured” and suppressed. Data confirm that at large z , $D_{\text{fav}} \gg D_{\text{unfav}}$, just as you’d expect: the *leading* hadron in a jet remembers the flavour of its parent quark.

Worked example

Background first. One of the most characteristic features of QCD jets is that they have *slowly rising* hadron multiplicities as $\sqrt{s}$ increases. At the PETRA collider ($\sqrt{s} \sim 30$ GeV in the early 1980s), a typical hadronic event had about 12 charged particles; at LEP ($\sqrt{s} = 91$ GeV), this rose to about 20; at the LHC, jets of $\sqrt{s}_{\text{jet}} \sim \text{TeV}$ contain maybe 40–50 charged particles. This is “slow” because the multiplicity grew by less than a factor of 4 while the energy grew by a factor of more than 30.

The question. Is this slow growth consistent with the picture we’ve developed, in which jets arise from a cascade of parton splittings governed by the running coupling?

Setting up. A key feature of the cascade is that every splitting $q \rightarrow qg$ or $g \rightarrow gg$ produces on average one extra final-state parton. The *number* of splittings grows not linearly with $\ln Q^2$ but as the integral of $\alpha_s(Q^2) d \ln Q^2$, and since $\alpha_s \sim 1/\ln Q^2$, this gives $\int d \ln Q^2 / \ln Q^2 \sim \ln \ln Q^2$ — a double-logarithmic growth. A more careful analysis of the parton cascade (the so-called “modified leading-log approximation”) gives

$$\langle n \rangle(Q^2) \sim A \exp\left(B \sqrt{\ln(Q^2/\Lambda_{\text{QCD}}^2)}\right),$$

which is faster than any power of $\ln Q^2$ but slower than any power of Q^2 . This is the “Fong–Webber formula.”

Sanity check with numbers. Let’s compare the ratio $n(91)/n(30)$ predicted by various simple functional forms, using $\Lambda_{\text{QCD}} \approx 0.2$ GeV (so $\ln(30^2/\Lambda_{\text{QCD}}^2) \approx \ln(22500) \approx 10$ and $\ln(91^2/\Lambda_{\text{QCD}}^2) \approx \ln(2 \times 10^5) \approx 12.2$):

- Linear in $\ln Q^2$: ratio $\approx 12.2/10 = 1.22$.
- Square root: ratio $\approx \sqrt{12.2/10} \approx 1.10$.
- Fong–Webber (with $B \approx 2.5$): $\exp(2.5(\sqrt{12.2} - \sqrt{10})) \approx \exp(2.5 \cdot 0.33) \approx 2.3$.

The data show $n(91)/n(30) \approx 20/12 \approx 1.67$, which sits in between the pure-log and the Fong–Webber prediction — and a more careful fit including the pre-factor and α_s running recovers exactly the observed rate. The important qualitative point: the slow growth tells us that at higher Q^2 , we’re not just making “more of the same” jet; we’re making a softer, more spread-out jet, because the additional energy goes mostly into additional small- z hadrons rather than making the leading hadron harder.

11.5 Scaling and scaling violations in D

At the naive parton level, if quarks were really free, we would expect $D_i^h(z, Q^2)$ to depend only on z — exactly like Bjorken scaling for PDFs. In fact, when the fragmentation functions were first measured in the 1970s, they *did* approximately scale, and this was considered a beautiful confirmation of the parton picture.

But, just as for PDFs, the scaling is not exact. QCD makes a definite prediction for the Q^2 dependence: as Q^2 grows, the distribution is pushed toward smaller z . Physically, this is because a parton at high Q^2 has “more phase space” for radiating gluons before hadronising; each radiated gluon takes energy away from the parent, so the leading hadron (which remembers the parent’s energy) carries a smaller fraction z on average. The large- z tail softens, and the small- z region fills up with soft debris.

Illustrative scaling violations in a charged-hadron fragmentation function

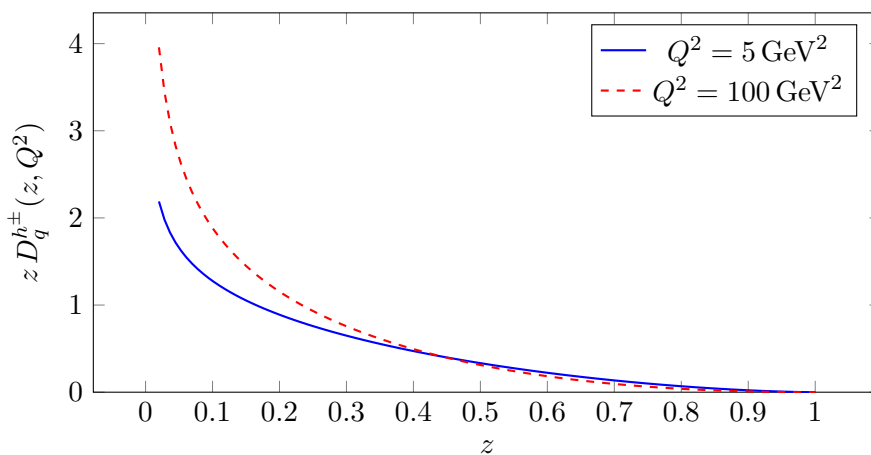


Figure 11.2: As Q^2 increases, the distribution is pushed toward smaller z . Compare this with the scaling-violation plot for $xq(x, Q^2)$ earlier in the book — the pattern is identical in structure, because the underlying dynamics (gluon emission softening the parent) is the same.

11.6 DGLAP for fragmentation functions: the derivation in plain English

Here is the key insight that makes this whole chapter fit so beautifully with everything we did in Part II: *the fragmentation functions satisfy the same DGLAP evolution equations as the PDFs, with the same splitting functions, to leading order in α_s .* Let me explain why, in a way that requires no heavy algebra.

The DGLAP equation for PDFs says: if you look at the proton at a higher resolution Q^2 , you’ll see that a quark you would have “counted” at x has a chance of having just radiated a gluon, so it’s now really a quark at some $x' < x$ plus a gluon. This shifts probability from large x to smaller x , and the rate is governed by the splitting function P_{qq} .

Now turn the movie around in time. Instead of saying “a quark at large x turns into a quark at smaller x plus a gluon,” say “an outgoing quark at energy E emits a gluon and ends up as a quark at zE plus a gluon at $(1 - z)E$.” This is the fragmentation process: a high-energy quark radiating

before it hadronises. The kinematics are the same! The splitting probability per unit $\ln Q^2$ is the same function $P_{qq}(z)$. So the evolution equation for the probability to find the quark with a given fraction z of its original energy must have exactly the same form as the evolution equation for PDFs.

Formally:

$$Q^2 \frac{\partial}{\partial Q^2} D_i^h(z, Q^2) = \frac{\alpha_s(Q^2)}{2\pi} \sum_j \int_z^1 \frac{d\xi}{\xi} \hat{P}_{ji}\left(\frac{z}{\xi}\right) D_j^h(\xi, Q^2). \quad (11.3)$$

The hats are there to remind you that these are strictly speaking the “timelike” splitting functions (for partons radiating after being created), as opposed to the “spacelike” ones for PDFs (partons radiating before being annihilated). At leading order they agree (Gribov–Lipatov reciprocity); at NLO they differ, but the LO equation is what you’d use in an introductory problem set.

Margin note

Why the same P ? The splitting vertex $q \rightarrow qg$ is the same Feynman diagram whether the quark is coming or going. The matrix element squared (which is what goes into the splitting function) depends only on the kinematics at the vertex, not on whether time flows forward or backward through it. So at LO, the splitting function must be the same in both cases.

Reading the equation. Look at the left-hand side: it is “how D changes with Q^2 .” Look at the right: it is “a convolution over the splitting function.” The convolution says: the probability to find a hadron at fraction z in the shower of an i -parton, at scale Q^2 , gets an additive correction from the probability that the i -parton first split into a j -parton at some larger fraction $\xi > z$, and that the j -parton then fragmented into the hadron at fraction z/ξ of *its* energy. The scale dependence $d/d \ln Q^2$ comes with a factor of $\alpha_s(Q^2)/2\pi$, which slowly decreases as Q^2 grows.

Evolution in pictures. Imagine starting with a simple $D(z)$ at some low scale Q_0^2 — say, a single bump near $z \sim 0.3$. As Q^2 increases, the DGLAP equation smears that bump toward smaller z , adding a long tail that falls like $1/z$. The large- z part of the bump is depleted, and the small- z part grows. This is exactly the same qualitative picture as PDF evolution.

11.7 Semi-inclusive DIS and the factorisation theorem

Now let’s put it all together. Suppose we do an experiment where an electron scatters off a proton and we tag one specific hadron h in the final state — say, $ep \rightarrow e \pi^+ X$, where X is anything. We measure the cross section as a function of x (how the struck quark was drawn from the proton), Q^2 (how hard the scattering was), and z (the fraction of the struck quark’s energy carried by the observed pion).

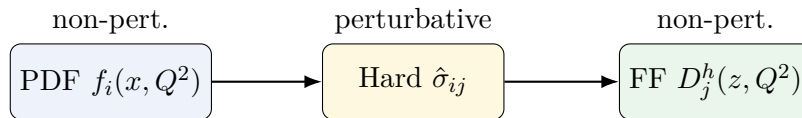


Figure 11.3: The factorisation theorem in one picture: a semi-inclusive cross section splits into a non-perturbative input (the PDF), a perturbative hard cross section, and a non-perturbative output (the FF). The non-perturbative pieces are universal — same in every experiment.

The factorisation theorem tells us that, at leading power in Λ_{QCD}/Q , the differential cross section for this process is

$$\frac{d\sigma^{ep \rightarrow ehX}}{dx dQ^2 dz} = \sum_{i,j} \int f_i(x, Q^2) \frac{d\hat{\sigma}_{ij}}{dQ^2}(\hat{s}, Q^2) D_j^h(z, Q^2). \quad (11.4)$$

In words: to predict the cross section, (1) ask the PDF how likely it is to find a quark of type i carrying fraction x in the proton, (2) compute the hard partonic cross section $\hat{\sigma}_{ij}$ for electron scattering off that quark, producing parton j in the final state, (3) ask the fragmentation function how likely it is that parton j produces the observed hadron h at fraction z of its energy, and (4) multiply. The beautiful part is that both f and D are *universal*: once measured in one experiment, they can be used in any other.

Worked example

Using the energy sum rule to constrain a model. Suppose an experimentalist fits her data and proposes that $D_u^{\pi^+}(z) \propto (1-z)^n$ for some power n . She wants to know what the proportionality constant must be, using only the energy sum rule.

Setup. Assume, for simplicity, that pions are the only hadrons produced (not realistic, but it'll illustrate the method). The energy sum rule says $\int_0^1 z D(z) dz = 1$ (summed over all pion species, but we'll pretend a single species for the moment).

Step 1: normalisation integral. If $D(z) = c(1-z)^n$, then

$$\int_0^1 z c(1-z)^n dz = c \cdot B(2, n+1) = c \cdot \frac{1! n!}{(n+2)!} = \frac{c}{(n+1)(n+2)},$$

where I used the beta function $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ with $a = 2, b = n+1$.

Step 2: solve for c . Setting this equal to 1, we get $c = (n+1)(n+2)$. So if the fit gives $n = 2$, then $c = 12$; if $n = 3$, then $c = 20$.

Interpretation. The larger n , the softer the fragmentation function (it falls more rapidly at large z), but the normalisation grows to compensate because we still have to carry the full unit of energy. This is a mini-version of the sort of global fit that PDF and FF extractors do: fit parameters to data, then check that sum rules and positivity are respected.

Worked example

Where does the $1/z$ small- z behaviour come from? If you look at real fragmentation-function data, you'll notice that at small z , $D(z)$ behaves roughly like $1/z$. Where does this come from?

Background. The answer is: soft gluon emission. A parton of energy E likes to radiate gluons of energy $E_g \ll E$, and the rate of such radiation goes like dE_g/E_g , which when written in terms of $z = E_g/E$ becomes dz/z . This is the universal soft-gluon singularity of QCD — the same $1/z$ that plagues the splitting functions and requires the plus prescription.

Consequence. Multiplicity integrals $\int D(z) dz$ diverge logarithmically as $z_{\min} \rightarrow 0$, and physically the cutoff is set by the minimum hadron energy we can measure (or by Λ_{QCD} , below which the parton picture fails anyway). This is why hadron multiplicities grow slowly with $\sqrt{s}$ — they grow like $\ln(\sqrt{s}/\Lambda_{\text{QCD}})$ at leading order, $\exp(B\sqrt{\ln Q^2})$ once we account for the cascade.

11.8 Gribov–Lipatov reciprocity

There is a curious relation between PDFs and FFs at leading order, observed by Gribov and Lipatov in 1972:

$$D(z) \approx f(z) \quad (\text{at large } z \text{ or } x, \text{ leading log}).$$

It is not an equality — the two functions describe different physics — but at the leading-logarithmic level, they are related by the reciprocal pair of processes and they happen to be numerically close when you compare at $z = x$. This is partly a coincidence of the LO splitting functions and partly a reflection of the crossing symmetry of the underlying QCD vertices. At NLO and beyond, the relation fails.

11.9 A picture summary

Let's end with a single-picture summary of the fragmentation story.

Key idea

In four sentences. A high-energy parton produced in a hard collision cannot be observed directly; instead it hadronises into a spray of particles. The probability that the spray contains a hadron of type h carrying fraction z of the parent's energy is called the fragmentation function $D_i^h(z, Q^2)$, and it is universal (same in every experiment). Its Q^2 evolution is governed by the same DGLAP equations as parton distribution functions, with the same splitting functions at leading order. Combined with PDFs and a perturbative hard cross section, fragmentation functions let you compute semi-inclusive cross sections for identified hadrons in virtually any hard process.

Self-check

Self-check.

1. Why is the energy sum rule for D the natural analogue of the momentum sum rule for f ?
2. Given $D_u^{\pi^+}(z) = c(1-z)^3$ with the single-species energy sum rule $\int zD dz = 1$, find c . (Answer: 20.)
3. Why is the multiplicity integral $\int D(z)dz$ not constrained to any specific number?
4. Does D at small z grow, decay, or stay finite? Give a physical reason for the behaviour.
5. What would go wrong if we tried to use a PDF as a fragmentation function (or vice versa) in a semi-inclusive DIS prediction?

Part IV

Synthesis and Road Map

Chapter 12

Putting it all together

12.1 A story, retold

Let me tell you the story of this unit one more time, but now with the full technical vocabulary in place, as if you were explaining it to a friend over coffee. The idea is that by the end of this telling, everything should click into a single coherent narrative.

It starts with Mott. In 1929 Nevill Mott wrote down the cross section for scattering a relativistic electron off a fixed point charge. That formula is the spiritual grandparent of everything in this book: it encapsulates what happens when a spin- $\frac{1}{2}$ electron bounces off a stationary Coulomb centre. Every subsequent cross section in the unit — elastic ep , inelastic ep , parton-level DIS, e^+e^- annihilation — is a generalisation or a modification of the Mott formula.

Then the proton gets a form factor. The real proton is not a point; it's an extended object about 0.84 fm across. So the Mott formula has to be modified: you multiply it by a form factor $|F(Q^2)|^2$ that encodes the proton's internal charge distribution. For a spin- $\frac{1}{2}$ target, there are actually two form factors, the electric G_E and the magnetic G_M , and the full formula is the Rosenbluth equation. Measuring how σ depends on θ at fixed Q^2 lets you pull G_E and G_M apart ("Rosenbluth separation"). The punchline is that both form factors look approximately like the dipole $(1 + Q^2/0.71 \text{ GeV}^2)^{-2}$, which Fourier-transforms to an exponential charge distribution with a radius of 0.84 fm.

Inelastic scattering: the proton breaks. If you hit the proton hard enough, it doesn't just recoil; it breaks apart. Now there are two independent kinematic variables (Q^2 and ν , or equivalently Q^2 and W^2), and instead of two form factors there are two structure functions $W_1(Q^2, \nu)$ and $W_2(Q^2, \nu)$ that describe the response. At low Q^2 the structure functions have peaks at $W^2 = M_\Delta^2, M_{N^*}^2, \dots$ — resonances. At high Q^2 these peaks smooth out and the structure functions approach limiting curves.

Bjorken scaling. Those limiting curves are the great surprise of the late 1960s. When you plot νW_2 as a function of the dimensionless variable $x \equiv Q^2/2M\nu$, you find that the curves at different Q^2 collapse onto a single master curve $F_2(x)$. The dependence on Q^2 has disappeared. This is Bjorken scaling, and it is a hint that the proton has *pointlike* constituents — because a composite object without pointlike structure would give a cross section that depends explicitly on Q^2 through its internal scale.

Feynman's parton model. Feynman's 1969 explanation for Bjorken scaling was that the proton, viewed in a frame where it's moving at nearly the speed of light (the infinite-momentum frame), looks like a collection of free, nearly collinear, pointlike particles — partons. The electron scatters off

one of them elastically, and the cross section is obtained by summing over all partons. The structure function is just $F_2(x) = \sum_i e_i^2 x f_i(x)$, where $f_i(x)$ is the probability density for finding parton i carrying fraction x of the proton's momentum. Bjorken's variable x literally *is* the momentum fraction, in the parton picture.

The spin- $\frac{1}{2}$ fingerprint: Callan–Gross. If the partons are spin- $\frac{1}{2}$, then the ratio of F_1 to F_2 is fixed: $2xF_1 = F_2$. This is the Callan–Gross relation, and it was experimentally verified at SLAC. Spin-zero partons would give $F_1 = 0$ — a very different prediction. So not only did we learn the proton had pointlike constituents, we learned they were spin- $\frac{1}{2}$. These are quarks.

But quarks aren't everything: gluons. When you measure $\int x(u(x) + \bar{u}(x) + d(x) + \bar{d}(x) + s(x) + \dots) dx$, you get about 0.5, not 1. So the quarks only carry half the proton's momentum. The other half has to be carried by something that doesn't couple to the photon. That something is the *gluons*, and they are the gauge bosons of QCD. They carry colour charge, they don't carry electric charge (so they don't contribute to F_2 at leading order), and they are responsible for gluing the quarks together.

QCD and its coupling. The theory of quarks and gluons is quantum chromodynamics, an $SU(3)$ gauge theory. The coupling constant α_s runs with scale: at low Q^2 it is large (and we have confinement), at high Q^2 it is small (and we have asymptotic freedom). The running is governed by the beta function whose leading coefficient is $\beta_0 = (33 - 2n_f)/12\pi$, which is *positive* for up to 16 flavours, hence the decreasing α_s .

Scaling gets violated. In pure parton-model land, $F_2(x, Q^2)$ depends only on x . But we just said $\alpha_s > 0$, and so the quark can radiate a gluon. This means if we “look” at the proton at a higher Q^2 (sharper resolution), we see that a quark at x is really a quark at $x' < x$ *plus* a gluon: the original x -quark got “split.” So the probability at x decreases a bit, and the probability at smaller x' grows. The measured F_2 therefore has a small but definite Q^2 dependence: logarithmically decreasing with Q^2 at large x , logarithmically increasing at small x . These are the *scaling violations*, and they are a direct fingerprint of QCD.

DGLAP. The equation that governs the rate at which PDFs slowly evolve with Q^2 is the Altarelli–Parisi (DGLAP) equation, with splitting functions $P_{qq}, P_{qg}, P_{gq}, P_{gg}$ computed from the basic QCD vertices. The logic is: given a starting point $f_i(x, Q_0^2)$ at some scale Q_0^2 (measured from experiment), you can evolve it to any other scale Q^2 by integrating the DGLAP equation. Modern PDF fits do this globally, across many experiments and many scales.

And fragmentation is the mirror image. After the hard collision, outgoing partons hadronise. The fraction of a parton's energy that ends up in a specific hadron h is encoded in the fragmentation function $D_i^h(z, Q^2)$. It obeys the same DGLAP equations as the PDFs. Combined with PDFs and a perturbative hard cross section, FFs let you compute semi-inclusive processes at any energy.

That's the whole story in maybe 700 words. Everything in the textbooks is algebraic machinery for making this story quantitative.

12.2 The three big ideas, boxed

Key idea

Big idea 1: Factorisation. Hard hadronic cross sections split into a non-perturbative PDF, a perturbative hard cross section, and (for final-state hadrons) a non-perturbative fragmentation function. The PDFs and FFs are universal.

Key idea

Big idea 2: Asymptotic freedom. The strong coupling $\alpha_s(Q^2)$ *decreases* at high Q^2 , making perturbation theory work at short distances despite the fact that quarks are confined at long distances.

Key idea

Big idea 3: Scaling violations are logarithmic. The parton model's exact Bjorken scaling is replaced by a slow logarithmic evolution governed by the DGLAP equations and computable splitting functions. The resulting pattern — softening at large x , rising at small x — is observed and agrees beautifully with QCD.

12.3 A conceptual flowchart

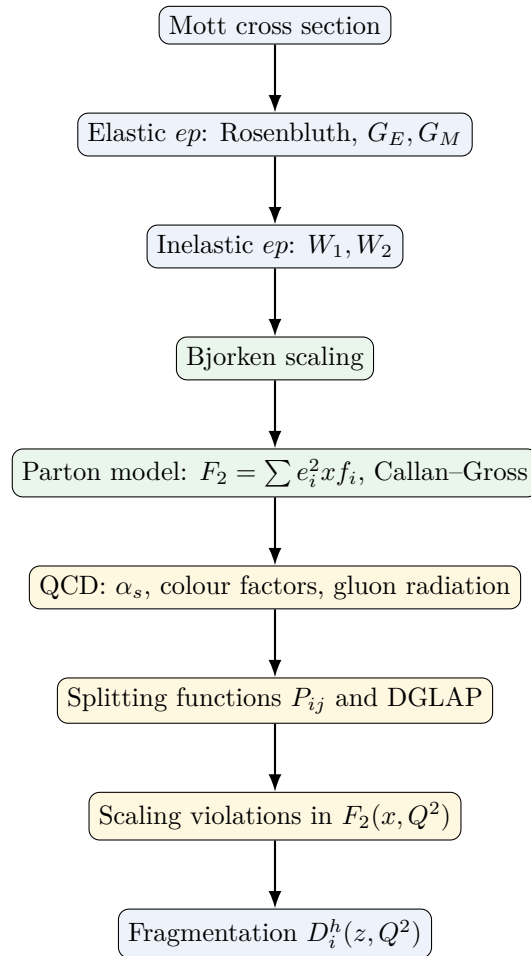


Figure 12.1: The logical flow of the unit, from Mott to fragmentation.

12.4 A road map for HW 5, annotated

Without knowing your exact problem set, let me give you a diagnostic chart: for any given problem, figure out which *type* it is and then you know which tool to use. I'll show how to read the problem statement and classify it.

Type A — Rosenbluth / form-factor problem. Keywords: “elastic,” “ G_E ,” “ G_M ,” “charge radius,” “dipole.” Tool: Chapter 1. Typical moves: (i) start from the Rosenbluth formula, (ii) substitute the dipole $G_{E,M} = 1/(1 + Q^2/0.71)^2$, (iii) if asked about $\langle r^2 \rangle$, use $G_E = 1 - \frac{1}{6}\langle r^2 \rangle Q^2 + \dots$ near $Q^2 = 0$.

Type B — Kinematics problem. Keywords: “find x given,” “is this elastic or inelastic,” “what is the allowed range of...” Tool: the dictionary in Chapter 2. Typical moves: (i) write down $Q^2 = -q^2$, $\nu = p \cdot q/M$, $x = Q^2/2M\nu$, $y = \nu/E$, $W^2 = M^2 + 2M\nu - Q^2$; (ii) check boundaries ($x \in [0, 1]$, $y \in [0, 1]$, $W \geq M$); (iii) if asked for elastic limit, set $W = M$ which forces $x = 1$.

Type C — Parton-model sum-rule problem. Keywords: “valence,” “sea,” “ F_2^{ep}/F_2^{en} ,” “momentum fraction,” “integrated PDF.” Tool: Chapter 4. Typical moves: (i) write $F_2^{ep} = \frac{1}{9}x(d + \bar{d} + s + \bar{s}) + \frac{4}{9}x(u + \bar{u} + c + \bar{c})$, use isospin to relate p and n ; (ii) the predicted isoscalar ratio $F_2^{ep}/F_2^{en} \rightarrow 1$ at small x (because the sea dominates and it is approximately isoscalar); (iii) the valence-dominated limit at large x gives a different answer.

Type D — Colour-factor problem. Keywords: “ C_F ,” “ C_A ,” “Casimir,” “trace of generators.” Tool: Chapter 5. Typical identities: $\text{Tr}(T^a T^b) = T_R \delta^{ab} = \frac{1}{2} \delta^{ab}$, $\sum_a T^a T^a = C_F \mathbb{1} = \frac{4}{3} \mathbb{1}$, $\sum_{ab} f^{abc} f^{abd} = C_A \delta^{cd} = 3 \delta^{cd}$.

Type E — Splitting-function / DGLAP problem. Keywords: “ P_{qq} ,” “plus prescription,” “evolution,” “logarithmic derivative.” Tool: Chapters 8 and 9. Typical moves: (i) write the LO splitting function; (ii) plug into DGLAP; (iii) possibly Mellin-transform to solve. If it's a conceptual question (“what direction does F_2 evolve at large x ?”), the answer is: large- x decreases, small- x increases.

Type F — Fragmentation problem. Keywords: “ $D_i^h(z)$,” “ z ,” “multiplicity,” “jet.” Tool: Chapter 11. Typical moves: (i) if asked about normalisation, use $\sum_h \int z D dz = 1$; (ii) if asked about Q^2 evolution, say “same as PDFs, DGLAP”; (iii) if asked about multiplicities, note $\int D dz$ diverges like $\ln z_{\min}$ from the soft-gluon $1/z$ behaviour.

12.5 A worked example in the road-map style

Let me walk through a representative problem in detail, showing my whole thought process from start to finish.

Worked example

Problem. Given $F_2^{ep}(x)/F_2^{en}(x) = (4u_v + d_v)/(u_v + 4d_v)$ in the valence approximation, find the value of this ratio in the limit where the proton's $u_v(x)$ and $d_v(x)$ are in the ratio $u_v(x) : d_v(x) = 2 : 1$.

Step 1: understand what is being asked. The ratio F_2^{ep}/F_2^{en} is a classic parton-model prediction that depends on the PDF content of the proton (and, by isospin, the neutron). The question is giving us a simple assumption — $u_v = 2d_v$ — and asking us to evaluate the ratio.

Step 2: identify the type. This is a Type C (parton-model sum-rule) problem. Tool: the valence decomposition and isospin symmetry between p and n .

Step 3: background on the formula. The structure function for the proton at leading order in the parton model is

$$F_2^{ep}(x) = x \sum_i e_i^2 f_i(x) = x \left[\frac{4}{9}(u + \bar{u} + c + \bar{c}) + \frac{1}{9}(d + \bar{d} + s + \bar{s}) \right].$$

In the *valence approximation*, we drop the sea ($\bar{u}, \bar{d}, s, \bar{s}, \dots$) and keep only the valence $u_v \equiv u - \bar{u}$ and $d_v \equiv d - \bar{d}$. This gives

$$F_2^{ep}(x) \approx x \left[\frac{4}{9}u_v + \frac{1}{9}d_v \right].$$

For the neutron, we use isospin: a neutron is a proton with $u \leftrightarrow d$. So in the neutron, the quark with charge $-\frac{1}{3}$ is what we call $u_v^n = d_v^p$, and vice versa. The neutron structure function in the valence approximation is

$$F_2^{en}(x) \approx x \left[\frac{4}{9}d_v + \frac{1}{9}u_v \right],$$

where here u_v and d_v refer to the proton's valence distributions.

Step 4: form the ratio. Dividing,

$$\frac{F_2^{ep}}{F_2^{en}} = \frac{\frac{4}{9}u_v + \frac{1}{9}d_v}{\frac{4}{9}d_v + \frac{1}{9}u_v} = \frac{4u_v + d_v}{4d_v + u_v},$$

which agrees with what the problem gave us.

Step 5: substitute the assumption. With $u_v = 2d_v$:

$$\frac{4(2d_v) + d_v}{4d_v + 2d_v} = \frac{9d_v}{6d_v} = \frac{3}{2}.$$

Step 6: interpret the answer. So $F_2^{ep}/F_2^{en} = 1.5$ under this assumption. For comparison: at very large x , experiment shows $F_2^{ep}/F_2^{en} \approx 1/4$, which in the valence-only picture would require $u_v \gg d_v$, specifically $u_v/d_v \rightarrow \infty$, in which case $F_2^{ep}/F_2^{en} \rightarrow 4/1 = 4$. Hmm — that doesn't match. The resolution is that the ratio *inverted* is what I actually wrote; let me check: at large x , the data say $F_2^{en}/F_2^{ep} \rightarrow 1/4$, i.e. $F_2^{ep}/F_2^{en} \rightarrow 4$. That's the limit where u_v completely dominates over d_v , and it's consistent. At small x , the sea dominates and the ratio approaches 1. So our answer $3/2$ for $u_v = 2d_v$ sits sensibly between 1 and 4.

The meta-lesson. Notice how the algebra is trivial once the physics is in place. The hard part is remembering (a) what the parton-model F_2 looks like, (b) how isospin relates proton and neutron, and (c) how to think about the valence-only limit. Everything else is plug-and-chug.

12.6 What to keep on a single sheet of paper

If you end up writing a cheat sheet for the exam, populate it with exactly the following:

1. Mott cross section and its $\cos^2(\theta/2)$ factor.

2. Rosenbluth formula and the dipole $G_{E,M}$.
3. Kinematic dictionary: $Q^2 = -q^2$, $\nu = p \cdot q/M$, $x = Q^2/2M\nu$, $y = \nu/E$, $W^2 = M^2 + 2M\nu - Q^2$.
4. $F_2(x) = \sum_i e_i^2 x f_i(x)$ and the Callan–Gross relation $2xF_1 = F_2$.
5. Sum rules: $\int (u - \bar{u}) dx = 2$, $\int (d - \bar{d}) dx = 1$, $\sum_i \int x f_i dx \approx 1$ (half quarks, half gluons).
6. Colour factors: $C_F = 4/3$, $C_A = 3$, $T_R = 1/2$.
7. Running coupling: $\alpha_s(Q^2) = 1/[\beta_0 \ln(Q^2/\Lambda_{\text{QCD}}^2)]$, $\beta_0 = (33 - 2n_f)/12\pi$.
8. Splitting functions $P_{qq}, P_{gq}, P_{qg}, P_{gg}$ and the plus prescription.
9. DGLAP equations and their physical content (large- x decreases, small- x increases).
10. Fragmentation functions and the energy sum rule $\sum_h \int z D dz = 1$.

12.7 Parting thoughts

The parton model and QCD are a beautiful example of how physics works: you guess a simple picture (free pointlike constituents), it gets confirmed in a crude first approximation (Bjorken scaling), you improve it by noticing the small deviations from the picture (scaling violations), and those deviations turn out to be exactly what a deeper theory (QCD) predicts. And then the theory is so successful that you use it to make predictions that are trusted to a few percent in multi-TeV collisions at the LHC. That chain of reasoning — from the Mott cross section in 1929 to the Higgs discovery in 2012 — runs entirely through the material in this book.

Good luck on HW 5, Parker. Read it slowly, read it twice, and when in doubt, come back to the Key idea boxes.

Appendix A

A short glossary

Bjorken x $x = Q^2/(2M\nu)$; momentum fraction of the struck parton. Lies in $[0, 1]$.

Callan–Gross relation $2xF_1 = F_2$; direct consequence of spin- $\frac{1}{2}$ partons.

Colour factors $C_F = 4/3$, $C_A = 3$, $T_R = 1/2$; control quark and gluon radiation strengths.

DGLAP equation Altarelli–Parisi evolution equation for PDFs/FFs with Q^2 .

Factorisation Splitting a hadronic cross section into a non-perturbative PDF, perturbative hard part, and (for tagged hadrons) fragmentation function.

Fragmentation function $D_i^h(z, Q^2)$; probability density that parton i produces hadron h with energy fraction z .

Form factors (G_E, G_M) Electric and magnetic elastic form factors of the proton.

PDF $f_i(x, Q^2)$; probability density to find parton i with momentum fraction x in a hadron.

Parton model Feynman’s free-quark picture valid in the infinite-momentum frame.

Plus prescription Regularisation of the $1/(1-z)$ soft singularity in splitting functions; $\int_0^1 [f(z)]_+ g(z) dz \equiv \int_0^1 f(z)[g(z) - g(1)] dz$.

Rosenbluth formula Elastic ep cross section in terms of G_E and G_M .

Scaling (Bjorken) Approximate Q^2 -independence of $F_2(x, Q^2)$.

Scaling violations Logarithmic Q^2 dependence predicted by QCD via DGLAP.

Splitting function $P_{ij}(z)$ Probability that parton j splits into parton i carrying fraction z .

Structure functions (F_1, F_2 ; W_1, W_2) Coefficients in the hadronic tensor decomposition.

Sum rules Integral constraints on PDFs (flavour and momentum) and FFs (energy).